



# ALPHA COLLEGE OF ENGINEERING

THIRUMAZHISAI, CHENNAI

(Approved by AICTE and Affiliated to Anna University)

## PROJECT DOCUMENTATION

### EventForce Management System

*Salesforce CRM Implementation for Event Planner Solutions*

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| Item          | Details                                                                                    |
|---------------|--------------------------------------------------------------------------------------------|
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## EVENTFORCE MANAGEMENT SYSTEM- SALESFORCE IMPLEMENTATION

### User Story:

The EventForce Management System is designed to provide a centralized Salesforce-based solution for managing event planning operations, including client bookings, event scheduling, venue reservations, vendor coordination, cancellation approvals, reminders, and feedback collection. The system enables event teams to manage Events, Clients, Vendors, Venues, and Feedback using a structured Salesforce data model with appropriate relationships. Automated Flows handle event reminders and process updates, while an Approval Process manages event cancellation requests. Apex Classes and Triggers support advanced business logic such as venue availability updates and prevention of double bookings. A Lightning App, Reports, and Dashboards provide real-time visibility into event schedules, vendor activities, and operational performance. This initiative improves event management efficiency, booking coordination, data accuracy, communication, operational visibility, and customer experience.

## 1. INTRODUCTION

### 1.1 Project Overview

The EventForce Management System is a Salesforce-based CRM solution developed to centralize and streamline event planning and management operations. The system manages clients, events, venues, vendors, feedback, and event cancellations within a structured Salesforce platform.

The solution uses custom objects, relationships, validation rules, Flows, Approval Processes, Apex Triggers and Classes, Reports, and Dashboards to automate key business processes. It supports venue availability management, prevention of double bookings, automated event reminders, cancellation approvals, feedback management, and operational reporting.

The project aims to reduce manual effort, improve booking coordination and data accuracy, enhance communication with clients and operational teams, and provide better visibility into event activities through a centralized Salesforce platform.

### 1.2 Purpose

The purpose of the EventForce Management System is to provide a centralized Salesforce platform for managing and automating event planning operations. The system is designed to:

- Centralize client, event, venue, vendor, and feedback information.
- Streamline event booking and venue reservation processes.
- Prevent double booking of venues through automated Apex logic.
- Automate event reminders and cancellation approval processes.
- Improve coordination between clients, event coordinators, vendors, and venue managers.
- Provide reports and dashboards for better operational visibility and decision-making.

- Reduce manual processes, improve data accuracy, and enhance the overall client experience.

**Expected Business Value**

| Business Outcome                  | Target                                                          |
|-----------------------------------|-----------------------------------------------------------------|
| Reduction in manual processes     | 60%                                                             |
| Improvement in booking efficiency | 40%                                                             |
| Client satisfaction               | Improved through timely reminders and communications            |
| Decision making                   | Data-driven insights through integrated reporting and analytics |

## 2. IDEATION PHASE

### 2.1 Problem Statement (Customer Problem Statement Template)

| PS    | I am (Customer)          | I'm trying to                                       | But                                                                                          | Because                                                                        | Which makes me feel                             |
|-------|--------------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------|
| PS -1 | Event Operations Manager | manage event planning operations efficiently        | client, event, venue, and vendor information is difficult to manage centrally                | event management activities require coordination across multiple areas         | frustrated and concerned                        |
| PS -2 | Event Coordinator        | create and manage event bookings efficiently        | event details, client information, and venue reservations require multiple manual activities | there is no centralized platform for managing the complete event lifecycle     | overwhelmed and frustrated                      |
| PS -3 | Venue Manager            | manage venue availability accurately                | the same venue may be assigned to multiple events                                            | venue bookings need proper availability tracking and double-booking prevention | concerned and uncertain                         |
| PS -4 | Vendor Manager           | coordinate vendors for different events             | vendor details and event assignments need to be managed separately                           | events can require multiple vendors and services                               | overwhelmed and concerned                       |
| PS -5 | Event Coordinator        | communicate event information and reminders on time | reminders have to be handled manually                                                        | there is a need to notify clients before upcoming events                       | stressed and concerned                          |
| PS -6 | Event Administrator      | manage event cancellation requests systematically   | cancellation requests require approval and status updates                                    | cancellation decisions need a controlled approval process                      | frustrated and uncertain                        |
| PS -7 | Event Coordinator        | collect and manage client feedback                  | feedback information needs to be maintained along with event and client details              | feedback is related to both events and clients                                 | concerned about customer satisfaction           |
| PS -8 | Event Operations Manager | monitor event operations and performance            | operational information is not easily visible in one place                                   | event data needs to be organized into reports and dashboards                   | concerned and less confident in decision-making |

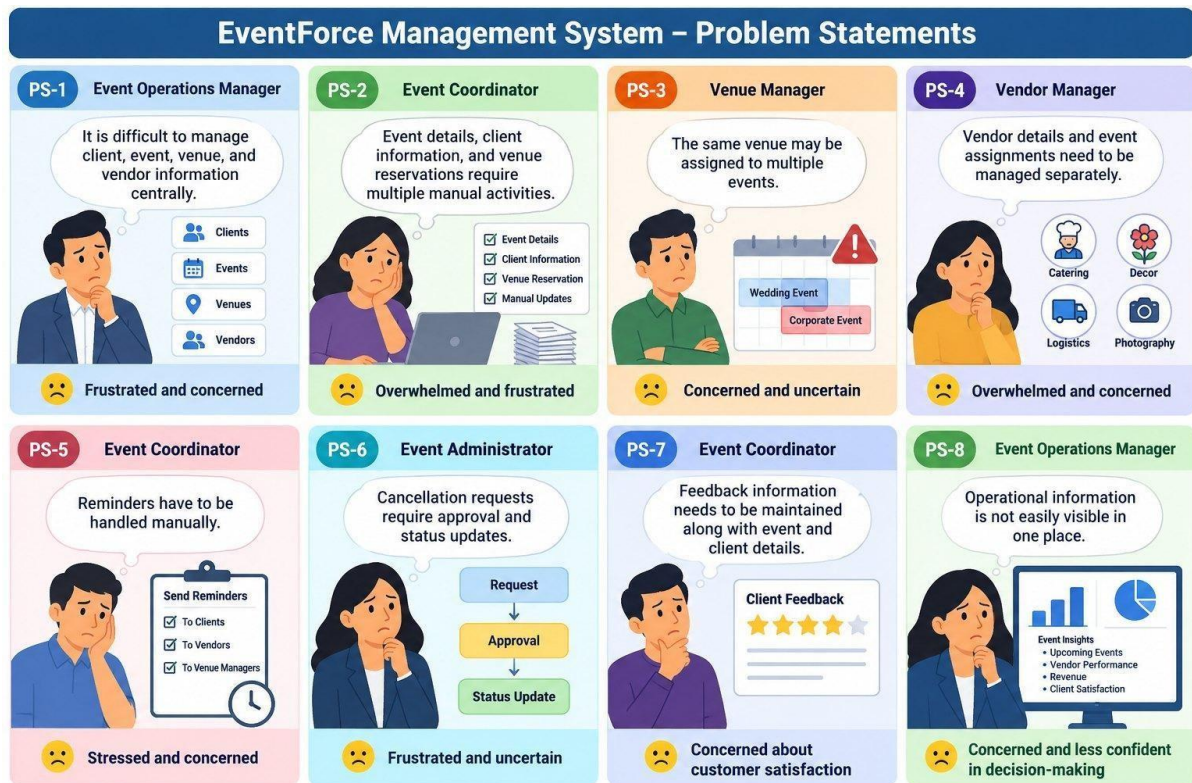


Figure 2.1 – Customer Problem Statements



## 2.2 Empathy Map Canvas

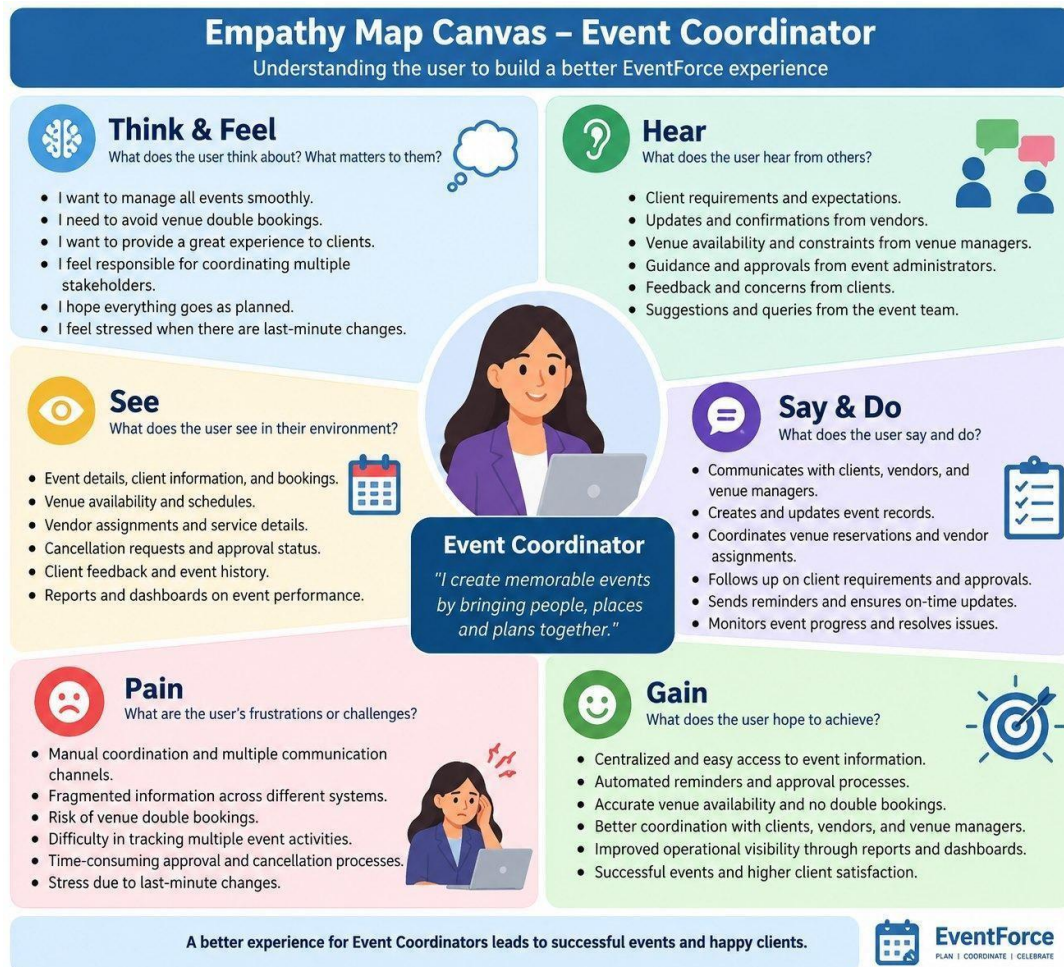


Figure 2.2 – Empathy Map Canvas: Event Coordinator

### User Persona: Event Coordinator

**Role:** Event Coordinator

**Primary Responsibility:** Manage event bookings, client requirements, venue reservations, vendor coordination, reminders, and event status updates.

- **Think & Feel:** Wants to manage events smoothly, avoid booking conflicts, and provide a reliable experience to clients.
- **Hear:** Receives requirements and updates from clients, vendors, venue managers, and event administrators.
- **See:** Event details, client information, venue availability, vendor assignments, cancellations, and feedback.
- **Say & Do:** Coordinates with stakeholders, updates event records, manages bookings, follows up on requirements, and monitors event progress.
- **Pain:** Manual coordination, fragmented information, venue double-booking risks, and difficulty tracking multiple event activities.
- **Gain:** Centralized event information, automated reminders and processes, accurate venue availability, better coordination, and improved operational visibility.

## 2.3 Brainstorming & Idea Prioritization

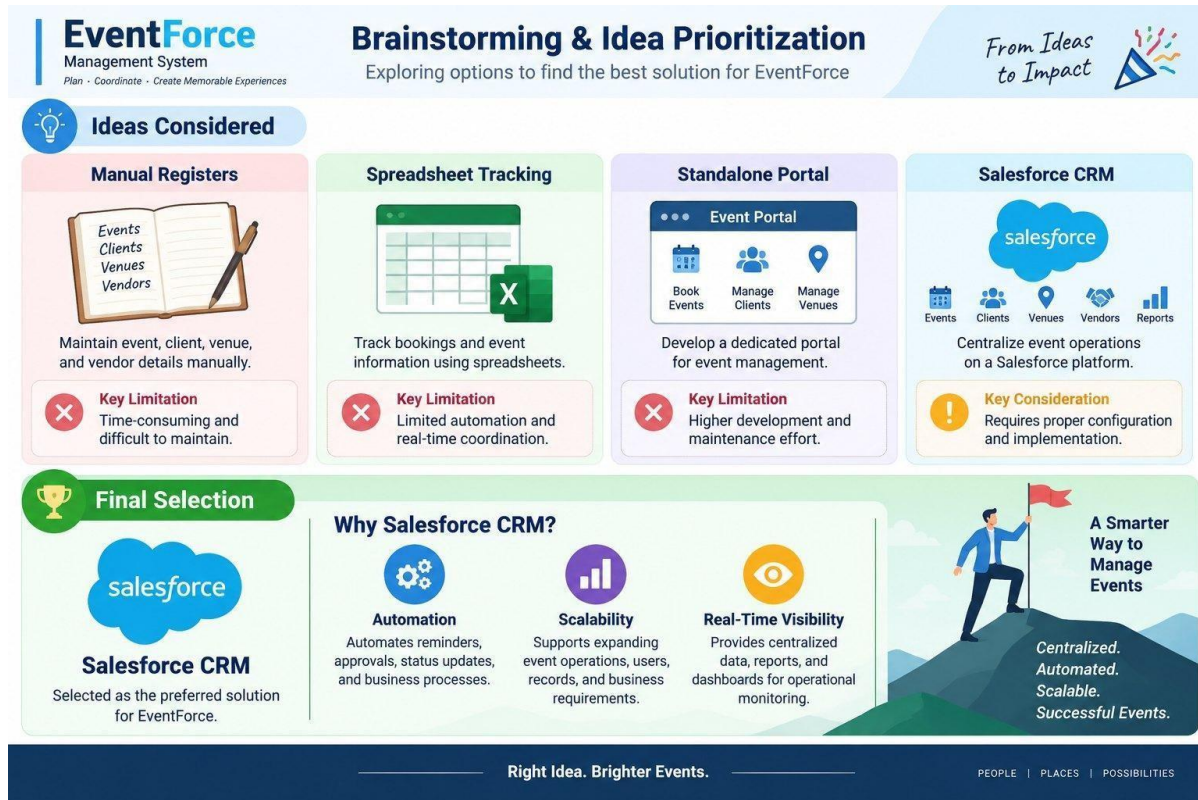


Figure 2.3 – Brainstorming & Idea Prioritization

### Ideas Considered

| Idea                 | Approach                                                   | Key Limitation                                   |
|----------------------|------------------------------------------------------------|--------------------------------------------------|
| Manual Registers     | Maintain event, client, venue, and vendor details manually | Time-consuming and difficult to maintain         |
| Spreadsheet Tracking | Track bookings and event information using spreadsheets    | Limited automation and real-time coordination    |
| Standalone Portal    | Develop a dedicated portal for event management            | Higher development and maintenance effort        |
| Salesforce CRM       | Centralize event operations on a Salesforce platform       | Requires proper configuration and implementation |

### Final Selection

Salesforce CRM selected due to:

- **Automation** – Automates reminders, approvals, status updates, and business processes.
- **Scalability** – Supports expanding event operations, users, records, and business requirements.
- **Real-Time Visibility** – Provides centralized data, reports, and dashboards for operational monitoring.

**Decision:** Salesforce CRM provides the best fit for creating a centralized, automated, and scalable EventForce Management System.



### 3. REQUIREMENT ANALYSIS PHASE

#### 3.1 Customer Journey Map

| Stage | Customer Journey Stage              | What Happens                                                                                                                |
|-------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 1     | Event Enquiry & Client Registration | Client requirements and contact details are captured in Salesforce.                                                         |
| 2     | Event Planning & Booking            | Event details such as event type, date, budget, and status are created and managed.                                         |
| 3     | Venue Selection & Reservation       | Suitable venues are identified, availability is checked, and reservations are managed while preventing double bookings.     |
| 4     | Vendor Coordination                 | Vendors and required services are assigned to events and their availability/status is coordinated.                          |
| 5     | Event Confirmation & Communication  | Confirmed events trigger automated communication, including reminders sent to clients before the event.                     |
| 6     | Event Execution & Monitoring        | Event progress and operational information are monitored through Salesforce records, reports, and dashboards.               |
| 7     | Cancellation & Approval             | Cancellation requests are submitted through the approval process, with appropriate status updates and notifications.        |
| 8     | Feedback & Closure                  | Client feedback is captured after the event, and completed events are updated for operational tracking and future analysis. |



Figure 3.1 – Customer Journey Map

## 3.2 Solution Requirements

### Functional Requirements

| FR No | Requirement                                    |
|-------|------------------------------------------------|
| FR-1  | Event management                               |
| FR-2  | Client management                              |
| FR-3  | Venue management                               |
| FR-4  | Vendor management                              |
| FR-5  | Event–vendor assignment                        |
| FR-6  | Venue availability & double-booking prevention |
| FR-7  | Automated event reminders                      |
| FR-8  | Event cancellation approval                    |
| FR-9  | Feedback management                            |
| FR-10 | Reports & dashboards                           |
| FR-11 | Event status automation                        |
| FR-12 | Data migration through Data Import Wizard      |

### Non-Functional Requirements

| NFR             | Description                                                              |
|-----------------|--------------------------------------------------------------------------|
| Usability       | Lightning App for easy event management and navigation                   |
| Security        | Profile-, role-, and permission-based access control                     |
| Reliability     | Accurate automation, approvals, and venue booking validation             |
| Performance     | Efficient processing of event and operational data                       |
| Availability    | Cloud-based Salesforce access for authorized users                       |
| Scalability     | Supports growing events, clients, vendors, venues, and users             |
| Maintainability | Flows, Apex, validation rules, and security can be monitored and updated |
| Data Integrity  | Validation and controlled data migration to maintain accurate records    |

### 3.3 Data Flow Diagram

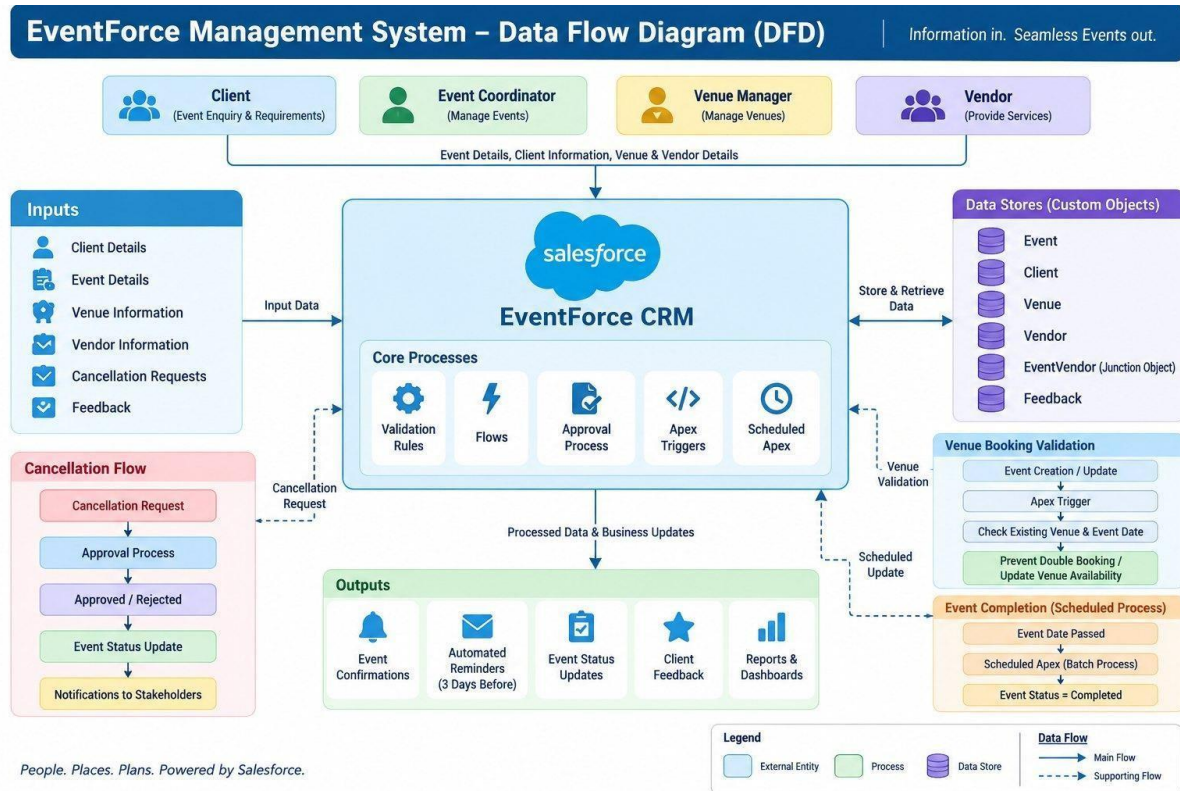
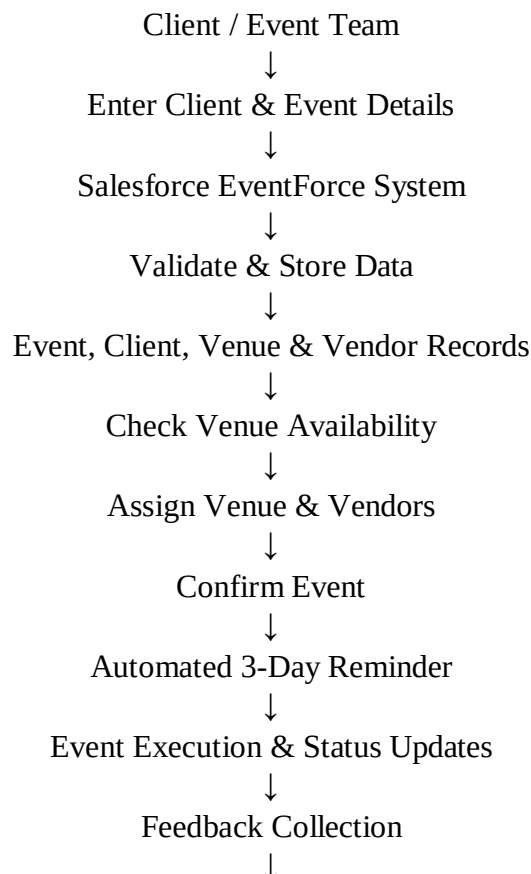


Figure 3.2 – Data Flow Diagram (DFD)

#### Flow:

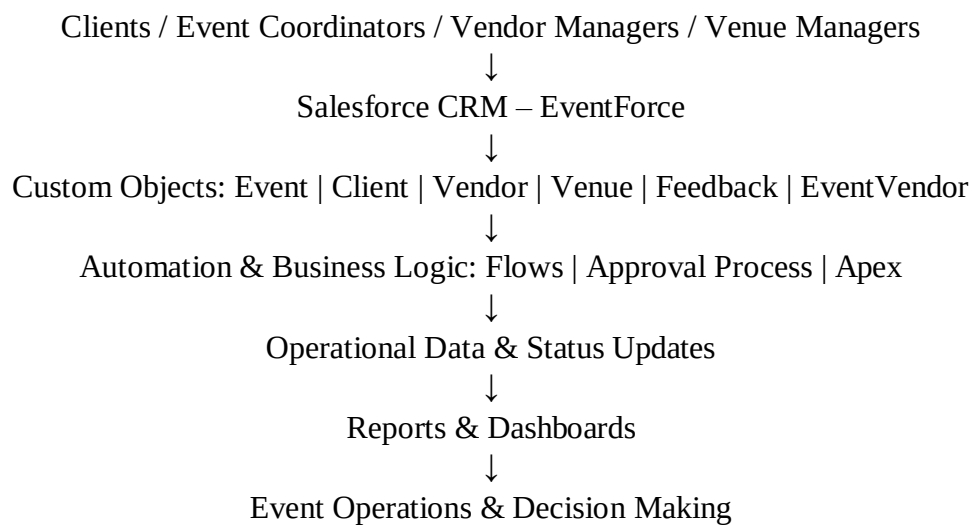


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## Supporting Flows

- **Cancellation Request:** Event Coordinator → Cancellation Request → Approval Process → Approved / Rejected → Event Status Update → Notifications
- **Venue Booking Validation:** Event → Apex Trigger → Venue Status Helper → Check Existing Venue & Event Date → Prevent Double Booking / Update Venue Availability
- **Event Completion:** Event Date Passed → Scheduled Apex → Batch Process → Event Status = Completed

## Simple DFD Representation



## 3.4 Technology Stack

| Layer          | Technology                                           |
|----------------|------------------------------------------------------|
| Platform       | Salesforce CRM / Developer Edition                   |
| UI             | Salesforce Lightning App                             |
| Database       | Salesforce Custom Objects                            |
| Business Logic | Salesforce Flows, Formula Fields, Validation Rules   |
| Approval       | Salesforce Approval Process                          |
| Automation     | Record-Triggered Flows, Apex Triggers                |
| Processing     | Batch Apex, Scheduled Apex                           |
| Security       | Profiles, Roles, Permission Sets, Sharing Rules, OWD |
| Reporting      | Reports & Dashboards                                 |



## 4. PROJECT DESIGN PHASE

### 4.1 Problem–Solution Fit

The EventForce Management System provides a strong fit between identified event-management problems and Salesforce capabilities. The solution combines custom data structures, automation, Apex business logic, approval processes, security controls, and reporting to address the operational challenges of event planning and coordination.

| Problem                                                                       | Solution – EventForce Salesforce System                                               | Fit / Benefit                                             |
|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Client, event, venue, and vendor information is difficult to manage centrally | Custom Salesforce objects for Event, Client, Venue, Vendor, Feedback, and EventVendor | Centralized and structured event data                     |
| Event booking requires multiple manual activities                             | Salesforce Flows and automated processes                                              | Reduces manual effort and improves process efficiency     |
| Risk of venue double booking                                                  | Apex Trigger and VenueStatusHelper                                                    | Validates venue availability and prevents double bookings |
| Vendor coordination is difficult to track                                     | EventVendor junction object and Vendor records                                        | Supports event–vendor assignment and coordination         |
| Clients may not receive timely event reminders                                | 3-Day Reminder Flow                                                                   | Automates reminder communication before confirmed events  |
| Cancellation requests require controlled processing                           | Event_Cancellation_Process Approval Process                                           | Provides structured approval and status updates           |
| Client feedback needs to be connected with events                             | Feedback object with Event and Client relationships                                   | Enables organized feedback management                     |
| Operational information is difficult to monitor                               | Reports and EventForce Operations Dashboard                                           | Provides better visibility for monitoring and decisions   |
| Different users require different levels of access                            | Profiles, Roles, Permission Sets, OWD, and Sharing Rules                              | Provides controlled and secure data access                |



Figure 4.1 – Problem–Solution Fit

## 4.2 Proposed Solution

| Parameter      | Description                                                             |
|----------------|-------------------------------------------------------------------------|
| Problem        | Manual and fragmented event management                                  |
| Solution       | Salesforce CRM                                                          |
| Innovation     | Automated event reminders, venue validation, and cancellation approvals |
| Impact         | Faster event coordination and improved operational visibility           |
| Business Model | Centralized event planning and management                               |
| Scalability    | Supports growing events, clients, vendors, venues, and users            |

## 4.3 Solution Architecture

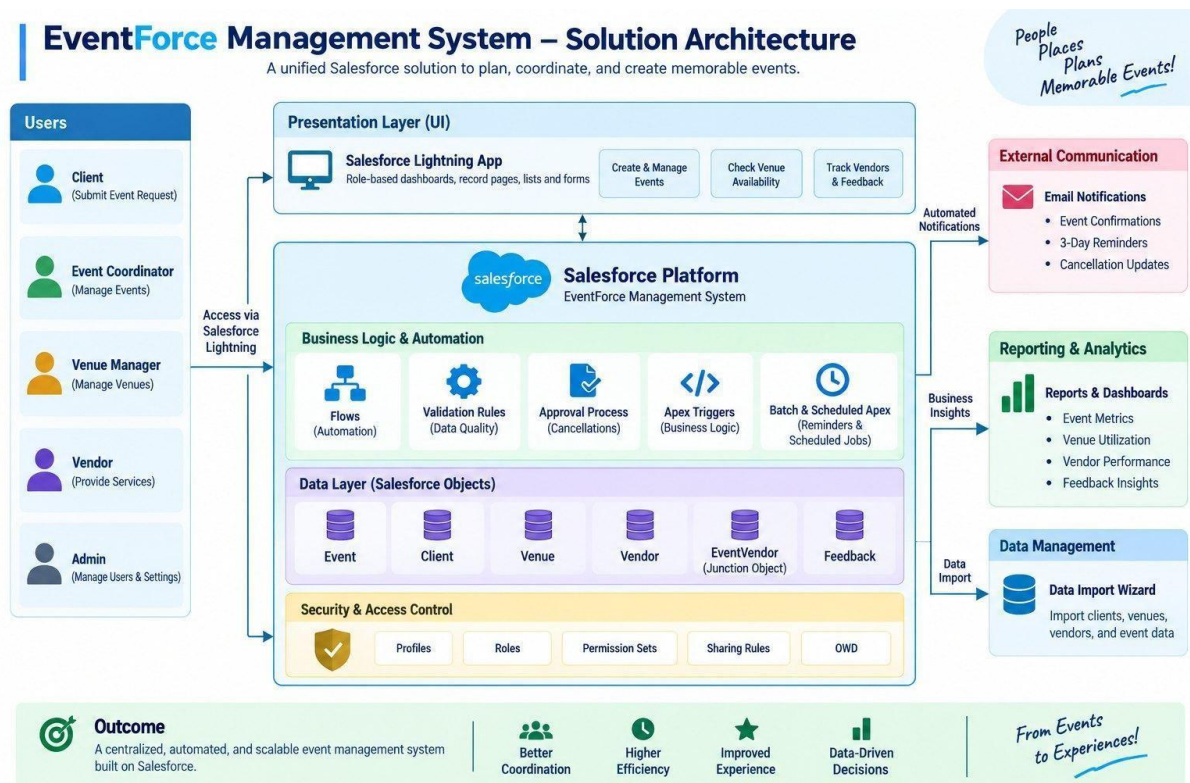


Figure 4.2 – Solution Architecture

### Components:

- Event Object
- Client Object
- Vendor Object
- Venue Object
- EventVendor Junction Object
- Feedback Object
- Validation Rules
- Formula Fields
- Flows
- Approval Process

- Apex Triggers

- Apex Classes
- Batch Apex
- Scheduled Apex
- Profiles & Roles
- Permission Sets & Sharing Rules
- Data Import Wizard
- Lightning App
- Reports
- Dashboards

## 5. PROJECT PLANNING & SCHEDULING

### 5.1 Project Planning

- Agile methodology
- Epics → Stories → Story Points
- Sprint-based execution
- Velocity calculation used for estimation

### Product Backlog, Sprint Schedule, and Estimation

| Sprint    | Functional Requirement (Epic) | User Story No. | User Story / Task                                                                                                         | Story Points | Priority | Team Member |
|-----------|-------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------|--------------|----------|-------------|
| Sprint -1 | Event Management              | US-01          | As an Event Coordinator, I want to create events with type, date, and status so that event information is standardized.   | 3            | High     | Pratiksha S |
| Sprint -1 | Client & Venue Management     | US-02          | As an Event Coordinator, I want to associate a client and venue with an event so that coordination is centralized.        | 5            | High     | Sowmiya A   |
| Sprint -2 | Vendor Management             | US-03          | As a Vendor Manager, I want to assign multiple vendors to an event so that services can be coordinated.                   | 5            | High     | Aswin S     |
| Sprint -2 | Notifications & Automation    | US-04          | As a Client, I want timely event reminders so that I know when my event is approaching.                                   | 3            | High     | Lokesh J    |
| Sprint -3 | Approval Management           | US-05          | As an Event Coordinator, I want cancellation requests to follow an approval process so that cancellations are controlled. | 5            | High     | Shalini A   |
| Sprint -3 | Venue Management & Apex       | US-06          | As a Venue Manager, I want double bookings prevented so that venue schedules remain accurate.                             | 8            | High     | Pratiksha A |
| Sprint -4 | Security & Access Control     | US-07          | As an Event Administrator, I want role-based access so that users see only appropriate information.                       | 5            | High     | Sowmiya A   |
| Sprint -4 | Reports & Dashboards          | US-08          | As Management, I want reports and dashboards so that event operations can be monitored.                                   | 5            | Medium   | Aswin S     |

### Project Tracker, Velocity & Burndown Chart



| Sprint   | Total Story Points | Duration | Sprint Start Date | Sprint End Date (Planned) | Story Points Completed | Sprint Release Date (Actual) |
|----------|--------------------|----------|-------------------|---------------------------|------------------------|------------------------------|
| Sprint-1 | 20                 | 6 Days   | 28 Sep 2026       | 03 Oct 2026               | 20                     | 29 Oct 2022                  |
| Sprint-2 | 20                 | 6 Days   | 31 Oct 2026       | 03 Oct 2026               |                        |                              |

| Sprint   | Total Story Points | Duration | Sprint Start Date | Sprint End Date (Planned) | Story Points Completed | Sprint Release Date (Actual) |
|----------|--------------------|----------|-------------------|---------------------------|------------------------|------------------------------|
| Sprint-3 | 20                 | 6 Days   | 27 Sep 2026       | 03 Oct 2026               |                        |                              |
| Sprint-4 | 20                 | 6 Days   | 2 Oct 2026        | 03 Oct 2026               |                        |                              |

Velocity: the team planned 20 story points per sprint (80 story points over 4 sprints of 6 days each), giving an average velocity of 20 story points per sprint, or about 3.33 story points per day ( $20 \div 6$ ).

## 6. PROJECT DEVELOPMENT PHASE

### 6.1 Phase 1: Requirement Analysis & Planning

Build an EventForce Management System in Salesforce to help event planners manage client bookings, vendor coordination, venue reservations, and feedback collection — all within a single CRM platform. The system automates key processes such as event scheduling, cancellation approvals, and client reminders, while providing real-time visibility through reports and dashboards.

This phase defined what the system must do, who will use it and in what order it would be built. The solution was scoped as a single Salesforce application that manages client bookings, vendor coordination, venue reservations and feedback, automates event scheduling, cancellation approvals and client reminders, and exposes real-time reports and dashboards.

#### 6.1.1 Functional Scope

- Custom objects: Event, Client, Vendor, Venue, Feedback and Event Vendor (Junction Object).
- Relationship modeling: Lookup, Master-Detail and Many-to-Many (Event ↔ Vendor).
- Automation using Flows for reminders, feedback notifications and approval updates.
- Validation Rules and Formula Fields for accurate data entry and event-budget calculation.
- Approval Processes for event cancellation requests with email alerts to coordinators and clients.
- Profiles, Roles and Permission Sets to manage user-level access and security.
- Reports and Dashboards for event summary, vendor performance and upcoming-event insights.

#### 6.1.2 Stakeholder Mapping

Stakeholders were mapped to Salesforce roles and profiles as shown below.

| Stakeholder         | Salesforce Role   | Salesforce Profile | Primary Needs                                     |
|---------------------|-------------------|--------------------|---------------------------------------------------|
| Event Administrator | Event Admin       | Event Admin        | Full system oversight, security and monitoring    |
| Event Coordinator   | Event Coordinator | Event Coordinator  | Event scheduling, client communication, approvals |
| Vendor Manager      | Vendor Manager    | Vendor Manager     | Vendor records and assigned services              |
| Client              | Client            | Client Profile     | View own events, submit feedback                  |

### 6.1.3 Execution Roadmap

| Step | Activity                          | Outcome                                                                 |
|------|-----------------------------------|-------------------------------------------------------------------------|
| 1    | Object & Field Setup              | Event, Client, Vendor, Venue, Feedback and Event Vendor objects created |
| 2    | Automation Flows                  | Record-triggered flow for event reminders                               |
| 3    | Validation Rules & Formula Fields | Data accuracy and budget calculations enforced                          |
| 4    | Profiles, Roles, Permission Sets  | User access and record visibility configured                            |
| 5    | Approval Process                  | Event cancellation workflow with notifications                          |
| 6    | Apex Classes & Triggers           | Venue availability updates and double-booking prevention                |
| 7    | Reports & Dashboards              | Analytics for upcoming events and performance                           |

## 6.2 Phase 2: Backend Development & Configurations

The backend phase built the functional engine of the system. Declarative tools (Record-Triggered Flows, Validation Rules, Formula Fields and Approval Processes) automated booking confirmations, reminders and cancellation approvals, while Apex classes and triggers handled advanced logic such as venue availability updates and double-booking prevention. The phase was executed through the milestones described below.

### 6.2.1 Milestone 1 – Salesforce Developer Account

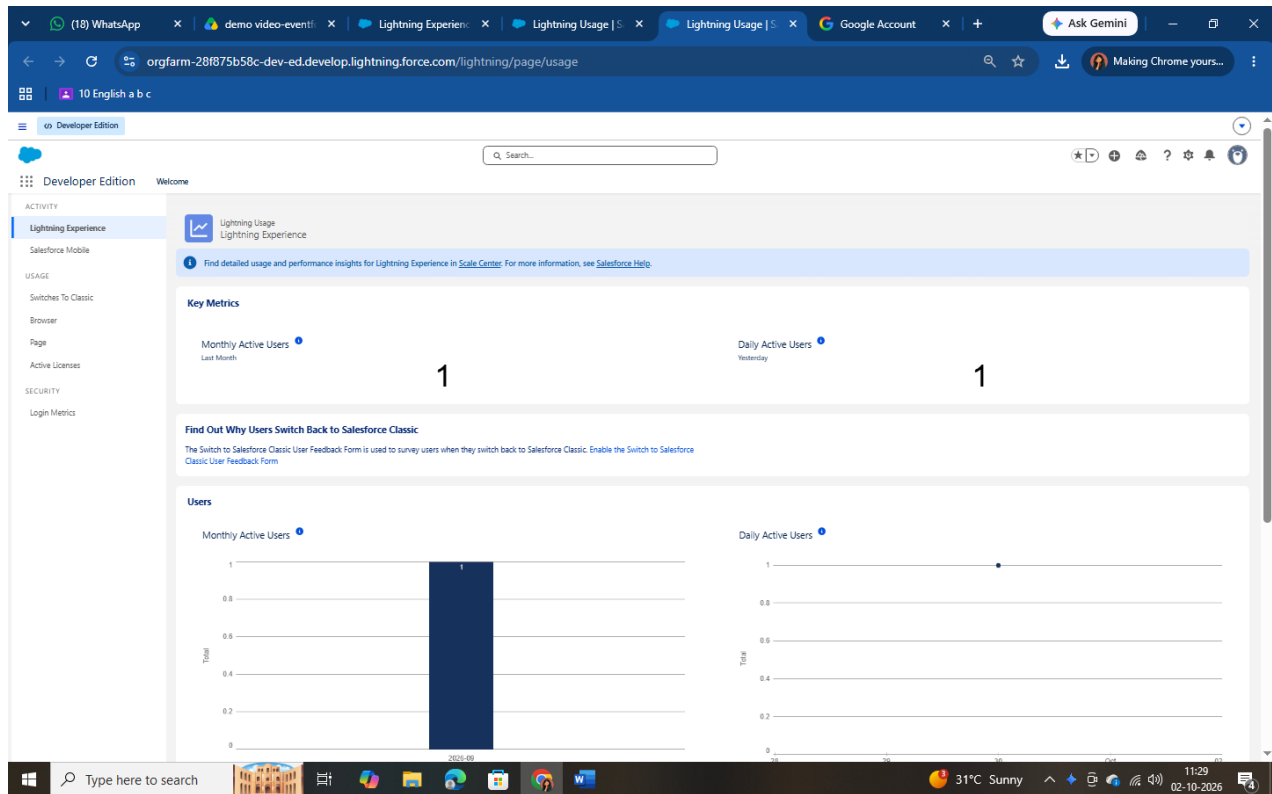
Salesforce is a customer success platform used to sell, service, market, analyze and connect with customers. A free Developer Edition org was created to build and test the entire project.

#### Activity 1: Creating the Developer Account

1. Open <https://developer.salesforce.com/signup>.
2. Enter First Name, Last Name and Email; select Role as Developer; enter the college name as Company; Country as India; and the Postal Code.
3. Enter a unique Username in email format (for example, username@organization.com). It need not be a real mailbox.
4. Click Sign me up.

#### Activity 2: Account Activation

1. Open the verification email (it may take 5–10 minutes) and click Verify Account.
2. Set a password and answer the security question, then click Change Password.
3. The user is redirected to the Salesforce Setup home page.



### 6.2.2 Data Architecture

The data model consists of five primary custom objects and one junction object. An object is a database table that stores organization-specific data; it consists of fields (columns) and records (rows).

| Object       | API Name       | Record Name                         | Purpose                                 |
|--------------|----------------|-------------------------------------|-----------------------------------------|
| Event        | Event__c       | Event Name (Text)                   | Stores event details, status and budget |
| Client       | Client__c      | Client Name (Text)                  | Stores client contact information       |
| Vendor       | Vendor__c      | Vendor Name (Text)                  | Stores vendor details and services      |
| Venue        | Venue__c       | Venue Name (Text)                   | Stores venue details and availability   |
| Feedback     | Feedback__c    | Feedback ID (Auto Number, F-{0000}) | Stores client ratings and comments      |
| Event Vendor | EventVendor__c | Auto-generated                      | Junction object linking Event ↔ Vendor  |

### Milestone 2 – Creating Custom Objects

Each object was created from Setup → Object Manager → Create → Custom Object. The Event object was created with the following options:

- Label: Event; Plural Label: Events; Record Name: Event Name; Data Type: Text.
- Optional Features: Allow Reports, Allow Activities and Track Field History.

- Deployment Status: Deployed; Search Status: Allow Search.
- Object Creation Options: Add Notes and Attachments related list to the default page layout.



The Client, Vendor and Venue objects were created in the same way (Record Name of Text type, with Allow Reports, Allow Search and the Notes and Attachments related list enabled). The Feedback object uses an Auto Number record name with display format F-{-0000} and a starting number of 0. The Event Vendor junction object was created separately (see Relationships under Section 6.2.2).

The top screenshot displays the Salesforce Object Manager interface with 233 items. The table below shows the objects listed:

| Object Name               | API Name                 | Object Type     | Created Date | Status | Actions |
|---------------------------|--------------------------|-----------------|--------------|--------|---------|
| Store                     | WebStore                 | Standard Object |              |        |         |
| Store Buyer Group         | WebStoreBuyerGroup       | Standard Object |              |        |         |
| Store Catalog             | WebStoreCatalog          | Standard Object |              |        |         |
| Task                      | Task                     | Standard Object |              |        |         |
| Time Slot                 | TimeSlot                 | Standard Object |              |        |         |
| UnstructuredStorageSpace  | UnstructuredStorageSpace | Standard Object |              |        |         |
| User                      | User                     | Standard Object |              |        |         |
| User Presence             | UserServicePresence      | Standard Object |              |        |         |
| User Provisioning Request | UserProvisioningRequest  | Standard Object |              |        |         |
| Utilization Alert         | TenantConsumptionAlert   | Standard Object |              |        |         |
| Vendor                    | Vendor__c                | Custom Object   | 9/30/2026    | ✓      | ⌵       |
| Vendor                    | EventVendor__c           | Custom Object   | 9/29/2026    | ✓      | ⌵       |
| Venue                     | Venue__c                 | Custom Object   | 9/30/2026    | ✓      | ⌵       |
| Voice Call                | VoiceCall                | Standard Object |              |        |         |
| Waitlist                  | Waitlist                 | Standard Object |              |        |         |
| Waitlist Participant      | WaitlistParticipant      | Standard Object |              |        |         |
| Waitlist Service Resource | WaitlistServiceResource  | Standard Object |              |        |         |
| Waitlist Work Type        | WaitlistWorkType         | Standard Object |              |        |         |

The bottom screenshot displays the Salesforce Object Manager interface with 160+ items. The table below shows the objects listed:

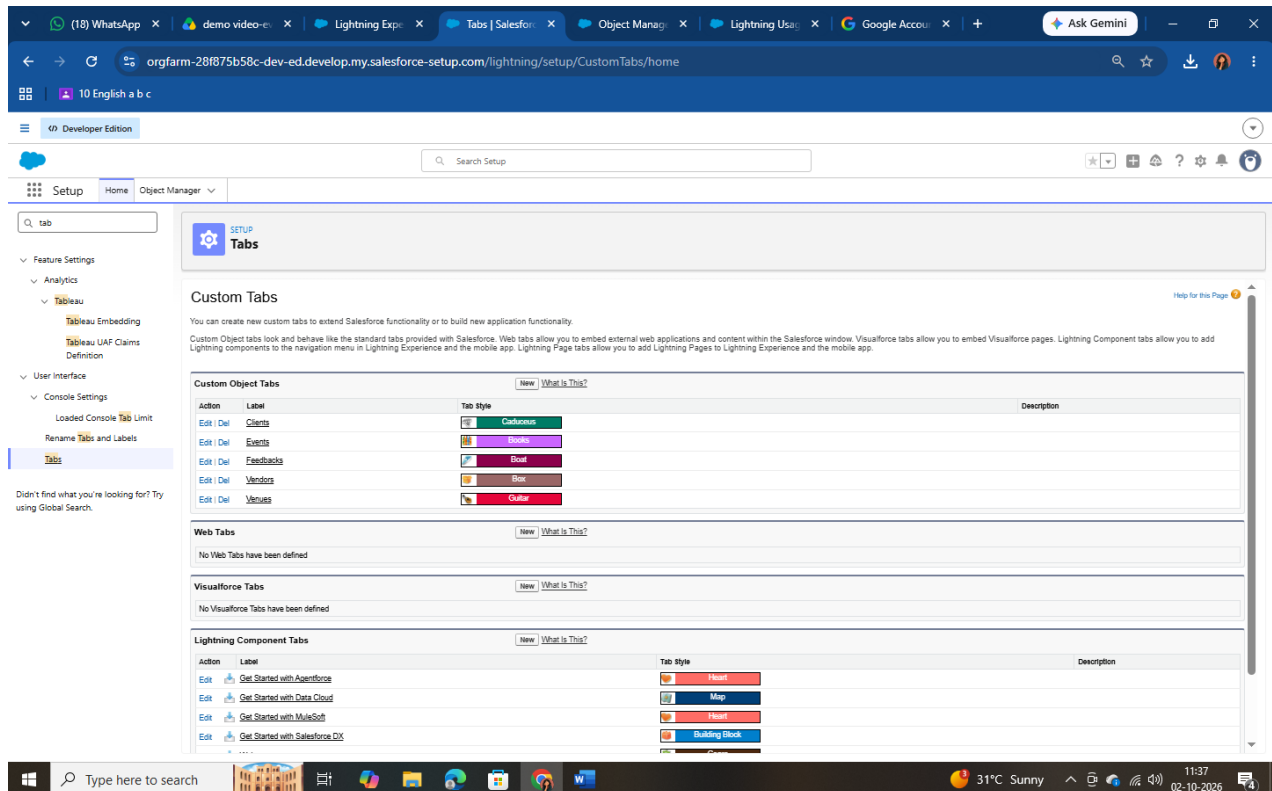
| Object Name                  | API Name                  | Object Type     | Created Date | Status | Actions |
|------------------------------|---------------------------|-----------------|--------------|--------|---------|
| Delivery Estimation Setup    | DeliveryEstimationSetup   | Standard Object |              |        |         |
| DevOps Pipeline              | DevOpsPipeline            | Standard Object |              |        |         |
| DevOps Project               | DevOpsProject             | Standard Object |              |        |         |
| Digital Wallet               | DigitalWallet             | Standard Object |              |        |         |
| Duplicate Record Item        | DuplicateRecordItem       | Standard Object |              |        |         |
| Duplicate Record Set         | DuplicateRecordSet        | Standard Object |              |        |         |
| Email Message                | EmailMessage              | Standard Object |              |        |         |
| Engagement Channel Type      | EngagementChannelType     | Standard Object |              |        |         |
| Engagement Channel Work Type | EngagementChannelWorkType | Standard Object |              |        |         |
| Entitlement                  | Entitlement               | Standard Object |              |        |         |
| Entitlement Contact          | EntitlementContact        | Standard Object |              |        |         |
| Event                        | Event__c                  | Custom Object   | 9/30/2026    | ✓      | ⌵       |
| Event                        | Event                     | Standard Object |              |        |         |
| Event Vendor                 | Event_Vendor__c           | Custom Object   | 9/30/2026    | ✓      | ⌵       |
| Feedback                     | Feedback__c               | Custom Object   | 10/1/2026    | ✓      | ⌵       |
| Finance Balance Snapshot     | FinanceBalanceSnapshot    | Standard Object |              |        |         |
| Finance Transaction          | FinanceTransaction        | Standard Object |              |        |         |
| Fulfillment Order            | FulfillmentOrder          | Standard Object |              |        |         |

The screenshot shows the Salesforce Object Manager interface. The browser address bar displays the URL: `orgfarm-28f875b58c-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ObjectManager/home`. The page title is "Object Manager" and it indicates "160+ items. Sorted by Label". A search bar labeled "Quick Find" is present, along with "Schema Builder" and "Create" buttons. The main content area is a table listing various objects:

| Object Name                             | API Name                    | Object Type     | Created Date | Status | Actions |
|-----------------------------------------|-----------------------------|-----------------|--------------|--------|---------|
| Cart Item Price Adjustment              | CartItemPriceAdjustment     | Standard Object |              |        |         |
| Cart Message                            | CartValidationOutput        | Standard Object |              |        |         |
| Cart Tax                                | CartTax                     | Standard Object |              |        |         |
| Case                                    | Case                        | Standard Object |              |        |         |
| Case Related Issue                      | CaseRelatedIssue            | Standard Object |              |        |         |
| Catalog                                 | ProductCatalog              | Standard Object |              |        |         |
| Category                                | ProductCategory             | Standard Object |              |        |         |
| Change Request                          | ChangeRequest               | Standard Object |              |        |         |
| Change Request Related Issue            | ChangeRequestRelatedIssue   | Standard Object |              |        |         |
| Change Request Related Item             | ChangeRequestRelatedItem    | Standard Object |              |        |         |
| Client                                  | Client_c                    | Custom Object   | 9/30/2028    | ✓      | ▼       |
| Communication Subscription              | CommSubscription            | Standard Object |              |        |         |
| Communication Subscription Channel Type | CommSubscriptionChannelType | Standard Object |              |        |         |
| Communication Subscription Consent      | CommSubscriptionConsent     | Standard Object |              |        |         |
| Communication Subscription Timing       | CommSubscriptionTiming      | Standard Object |              |        |         |
| Consumption Rate                        | ConsumptionRate             | Standard Object |              |        |         |
| Consumption Schedule                    | ConsumptionSchedule         | Standard Object |              |        |         |
| Contact                                 | Contact                     | Standard Object |              |        |         |
| Contact Point Address                   | ContactPointAddress         | Standard Object |              |        |         |

### Milestone 3 – Custom Tabs

Tabs display object data at a glance. A custom object tab was created for each object using Setup → Tabs → Custom Object Tabs → New. For each of Event, Client, Vendor, Venue and Feedback: select the object, choose a Tab Style icon, and accept the defaults (Next, Next, Save).



## Milestone 4 – Fields

Custom fields were added to each object from Object Manager → Fields & Relationships → New, choosing the appropriate data type.

| Object | Field Label  | Data Type          | Values / Notes                                                                                                                       |
|--------|--------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Event  | Event Type   | Picklist           | Wedding, Corporate, Birthday, Anniversary, Festival, Concert, Other                                                                  |
| Event  | Event Date   | Date               | Date of the event                                                                                                                    |
| Event  | Event Status | Picklist           | Planned, Confirmed, Completed, Pending Cancellation, Canceled, Rejected                                                              |
| Event  | Event Budget | Formula (Currency) | Calculated from Event Type (see Section 6.2.3)                                                                                       |
| Client | Email        | Email              | Client email address                                                                                                                 |
| Client | Phone        | Phone              | Client contact number                                                                                                                |
| Client | Address      | Text Area          | Client address                                                                                                                       |
| Client | Country      | Picklist           | India, Philippines, Saudi Arabia, USA, Switzerland, Thailand, Maldives, Dubai, Sri Lanka, Bangkok, Singapore, Myanmar, France, Nepal |

|        |              |                       |                                                                                                                                                                                                                                        |
|--------|--------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Client | City         | Picklist              | Hyderabad, Mumbai, Delhi, Bangalore, Chennai, Alaminos, Angeles City, Palayan, Quezon City, San Juan, Riyadh, Jeddah, Tayma, Dammam, Medina, New York, Los Angeles, Chicago, Houston, Columbus, Geneva, Basel, Lausanne, Bern, Lucerne |
| Vendor | Service Type | Multi-Select Picklist | Catering, Decor, Photography, Videography, Lighting, Stage Setup,                                                                                                                                                                      |

| Object   | Field Label         | Data Type      | Values / Notes                                          |
|----------|---------------------|----------------|---------------------------------------------------------|
|          |                     |                | Makeup Artist, DJ/Music, Transportation, Hosting/Anchor |
| Vendor   | Email               | Email          | Vendor email address                                    |
| Vendor   | Phone               | Phone          | Vendor contact number                                   |
| Vendor   | Status              | Picklist       | Available, Booked, Cancelled                            |
| Venue    | Address             | Text Area      | Venue address                                           |
| Venue    | Location            | URL            | Map link of the venue                                   |
| Venue    | Capacity            | Number         | Maximum guest capacity                                  |
| Venue    | Availability Status | Picklist       | Available, Reserved, Not Available                      |
| Feedback | Rating              | Picklist       | 1, 2, 3, 4, 5                                           |
| Feedback | Comments            | Long Text Area | Client comments                                         |

The screenshot shows the Salesforce Setup interface for the 'Event' object. The 'Fields & Relationships' section is active, displaying a table of fields. The table has columns for Field Label, Field Name, Data Type, Controlling Field, and Indexed. The fields listed are:

| Field Label      | Field Name       | Data Type          | Controlling Field | Indexed |
|------------------|------------------|--------------------|-------------------|---------|
| Client           | Client__c        | Lookup(Client)     |                   | ✓       |
| Created By       | CreatedById      | Lookup(User)       |                   |         |
| Event Budget     | Event_Budget__c  | Formula (Currency) |                   |         |
| Event Date       | Event_Date__c    | Date               |                   |         |
| Event Name       | Name             | Text(80)           |                   | ✓       |
| Event Status     | Event_Status__c  | Picklist           |                   |         |
| Event Type       | Event_Type__c    | Picklist           |                   |         |
| Last Modified By | LastModifiedById | Lookup(User)       |                   |         |
| Owner            | OwnerId          | Lookup(User:Group) |                   | ✓       |
| Venue            | Venue__c         | Lookup(Venue)      |                   | ✓       |

## Relationships

Relationships connect the objects. Lookup relationships link loosely coupled records, whereas Master-Detail relationships create a tight parent–child dependency. Salesforce does not allow a direct many-to-many relationship, so the Event Vendor junction object with two Master-Detail relationships was created to link Event and Vendor.

| Relationship   | Type   | Field Label |
|----------------|--------|-------------|
| Event → Client | Lookup | Client      |
| Event → Venue  | Lookup | Venue       |

| Relationship         | Type          | Field Label |
|----------------------|---------------|-------------|
| Feedback → Event     | Lookup        | Event       |
| Feedback → Client    | Lookup        | Client      |
| EventVendor → Event  | Master-Detail | Event       |
| EventVendor → Vendor | Master-Detail | Vendor      |

### Creating the Lookup Relationships

1. From Setup open Object Manager and select the Event object.
2. Click Fields & Relationships → New → Lookup Relationship → Next.
3. Choose the related object (Venue), click Next, enter the field label and continue with Next, Next, Next and Save.
4. Repeat to add the Client lookup on Event, and the Event and Client lookups on Feedback.

### Creating the Many-to-Many Relationship (Event ↔ Vendor)

1. Go to Object Manager → Create → Custom Object. Label: Event Vendor; Object Name: EventVendor\_\_c.
2. On the Event Vendor object click Fields & Relationships → New → Master-Detail Relationship. Choose Event as the related object, label it Event and click Save & New.
3. Create a second Master-Detail Relationship to Vendor with the label Vendor and click Save.

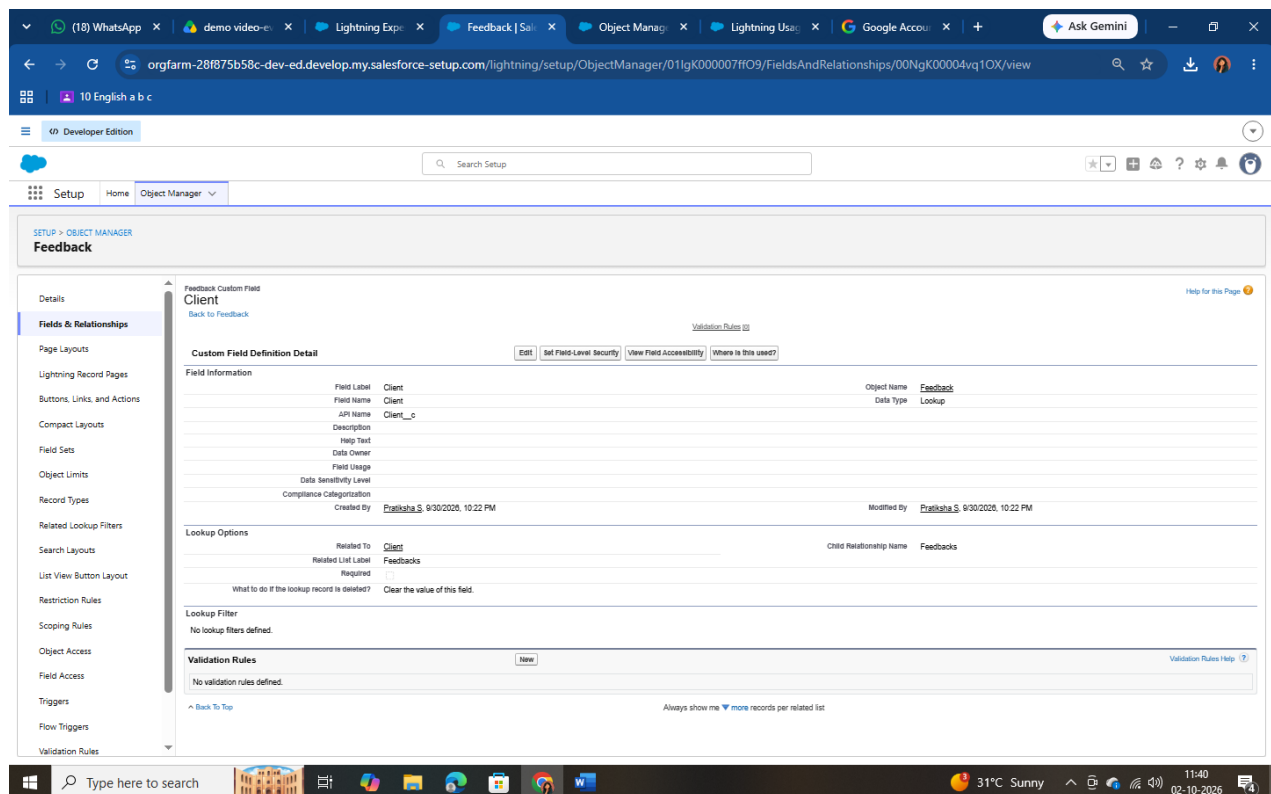
The screenshot shows the Salesforce Object Manager interface for the 'Event' object. The 'Fields & Relationships' section is active, displaying a table of fields. The table has columns for Field Label, Field Name, Data Type, Controlling Field, and Indexed. The fields listed are:

| FIELD LABEL      | FIELD NAME       | DATA TYPE          | CONTROLLING FIELD | INDEXED |
|------------------|------------------|--------------------|-------------------|---------|
| Client           | Client__c        | Lookup(Client)     |                   | ✓       |
| Created By       | CreatedById      | Lookup(User)       |                   |         |
| Event Budget     | Event_Budget__c  | Formula (Currency) |                   |         |
| Event Date       | Event_Date__c    | Date               |                   |         |
| Event Name       | Name             | Text(80)           |                   | ✓       |
| Event Status     | Event_Status__c  | Picklist           |                   |         |
| Event Type       | Event_Type__c    | Picklist           |                   |         |
| Last Modified By | LastModifiedById | Lookup(User)       |                   |         |
| Owner            | OwnerId          | Lookup(User/Group) |                   | ✓       |
| Venue            | Venue__c         | Lookup(Venue)      |                   | ✓       |

## Lookup Filter (Feedback → Event)

Scenario: when a Client is selected on a Feedback record, the Event lookup must show only that client's events. For example, if Ramesh is selected, only Ramesh's events should appear.

1. Open Object Manager → Feedback → Fields & Relationships → Event (Event\_\_c) and click Edit.
2. In the Lookup Filter section click Show Filter Settings.
3. Build the criteria: Event: Client (Client\_\_c) equals Field → Feedback: Client (Client\_\_c).
4. Tick Enable this filter and click Save.



## 6.2.3 Formula Field – Event Budget

The Event Budget is a Formula field (return type Currency) on the Event object that derives the budget automatically from the Event Type, removing manual entry errors. Steps: Fields & Relationships → New → Formula → Label “Event Budget” → Return type Currency → insert the formula below → Check Syntax → Next, Next, Save.

```
CASE (
  TEXT(Event_Type__c),
  "Wedding", 50000,
  "Corporate", 30000,
  "Birthday", 10000,
  "Anniversary", 20000,
  "Festival", 60000,
  "Concert", 40000,
  "Other", 15000,
  0
)
```



| Event Type | Budget |
|------------|--------|
| Wedding    | 50,000 |

| Event Type  | Budget |
|-------------|--------|
| Corporate   | 30,000 |
| Birthday    | 10,000 |
| Anniversary | 20,000 |
| Festival    | 60,000 |
| Concert     | 40,000 |
| Other       | 15,000 |

The screenshot shows the Salesforce Setup interface. The left sidebar lists various setup options, and the main content area displays the 'Event Custom Field' definition for 'Event Budget'. The field is of type 'Text' with a length of 255 characters. The formula options section shows a CASE statement that maps event types to their respective budget values.

| Field Label               | Event Budget                     | Object Name |
|---------------------------|----------------------------------|-------------|
| Field Name                | Event_Budget__c                  | Event       |
| API Name                  | Event_Budget__c                  |             |
| Description               |                                  |             |
| Help Text                 |                                  |             |
| Data Owner                |                                  |             |
| Field Usage               |                                  |             |
| Data Sensitivity Level    |                                  |             |
| Compliance Categorization |                                  |             |
| Created By                | Pratiksha S. 9/30/2026, 11:01 PM |             |
| Modified By               | Pratiksha S. 9/30/2026, 11:01 PM |             |

**Formula Options**

| Data Type      | Formula |
|----------------|---------|
| Decimal Places | 2       |

**Formula:**

```

CASE(
  TEXT(Event_Type__c),
  "Wedding", 50000,
  "Corporate", 30000,
  "Birthday", 10000,
  "Anniversary", 20000,
  "Festival", 60000,
  "Concert", 40000,
  "Other", 15000,
  0
)

```

## 6.2.4 Milestone 5 – Validation Rules

Validation rules run when a user saves a record and block the save if the data does not meet the defined criteria. A validation rule was created on the Client object to ensure that only properly formatted email addresses are stored.

1. Go to Setup → Object Manager → Client → Validation Rules → New.
2. Rule Name: Email\_Valid\_Address; tick Active.
3. Enter the Error Condition Formula shown below.
4. Error Message: "Please Enter Valid Email Address"; Error Location: Field → Email.  
Click Save.

`NOT(REGEX(Email__c, "^[a-zA-Z0-9._]+@[a-zA-Z0-9]+\.[a-zA-Z]{2,}$"))`

### 6.2.5 Milestone 6 – Approval Process (Event Cancellation)

## Step A – Classic Email Templates

| Template Name                           | Subject                                         | Used For                                         |
|-----------------------------------------|-------------------------------------------------|--------------------------------------------------|
| Event Cancellation Request Notification | Approval Request: Cancel Event {!Event__c.Name} | Approval request submission alert                |
| Email to Client After Cancellation      | Event Cancellation Notice – {!Event__c.Name}    | Informs client once the cancellation is approved |
| Approval Notification to Event Owner    | Event Cancellation Approved: {!Event__c.Name}   | Owner notification on approval                   |
| Rejection Notification to Event Owner   | Event Cancellation Rejected: {!Event__c.Name}   | Owner notification on rejection                  |

### Sample Template Body – Event Cancellation Request Notification

Hello!

An approval request has been submitted to cancel the event:

- Event Name: {!Event\_\_c.Name}
- Event Date: {!Event\_\_c.Event\_Date\_\_c}

Thank you,

EventForce Management System

### **Sample Template Body – Approval Notification to Event Owner**

Hello {!Event\_\_c.OwnerFullName},

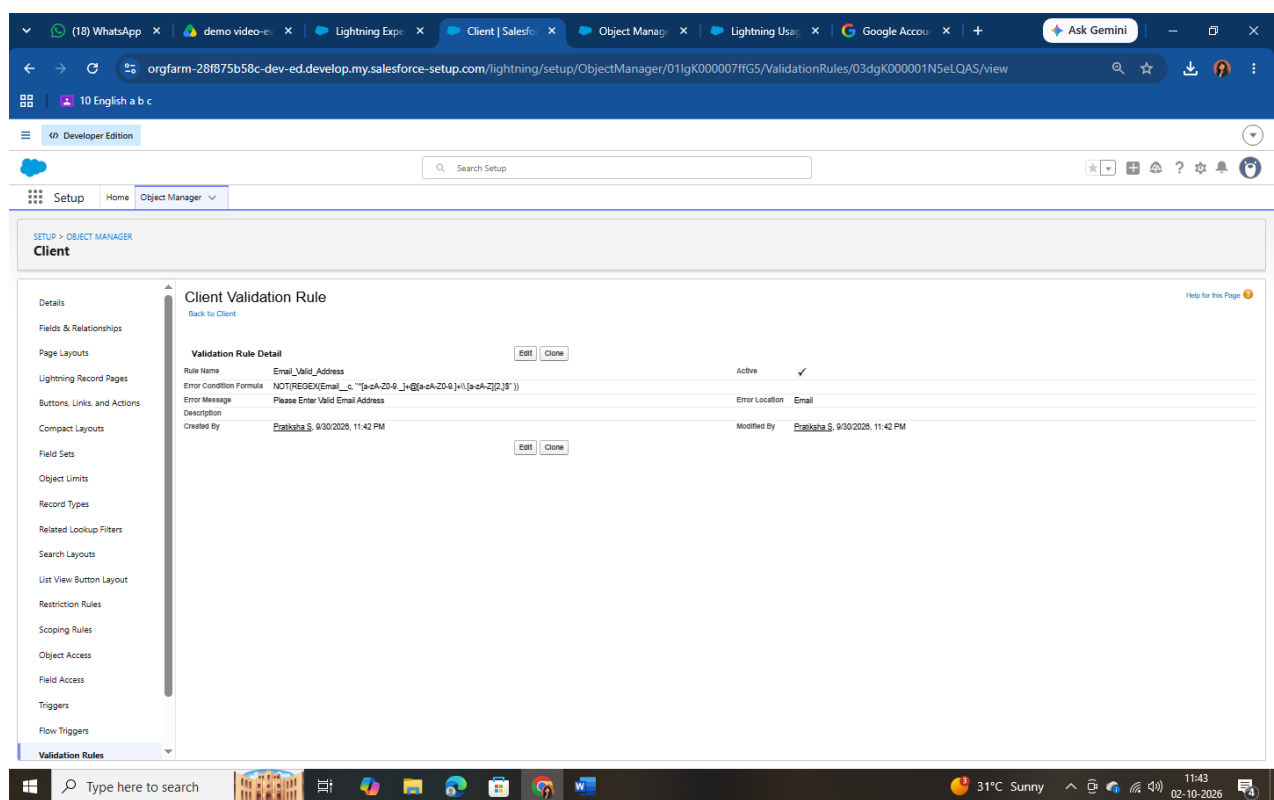
Your cancellation request for the event {!Event\_\_c.Name} has been approved.

Event Date: {!Event\_\_c.Event\_Date\_\_c}

The event status has been updated to Canceled.

Thank you,  
EventForce Management System

The Rejection Notification template follows the same structure, stating that the request has been rejected and that the event status has been reverted to Confirmed. The Email to Client After Cancellation template informs the client that the event (name and date) has been canceled and invites them to contact the team with any questions.



## Step B – Creating the Approval Process

1. Setup → Approval Processes → select object Event → Create New Approval Process → Use Standard Setup Wizard.
2. Basic Settings: Name = Event\_Cancellation\_Process; unique name auto-fills.
3. Entry Criteria: Criteria are met → Event: Event Status equals Pending Cancellation.
4. Select Approver: automatically assign using a user or queue (a specific user / manager of the submitting user; ensure a manager is set on the User record).
5. Email Template: Event Cancellation Request Notification.
6. Approval Page Fields: Event Name, Owner, Client, Last Modified By, Created By, Event Budget, Event Date, Event Status, Event Type, Venue.
7. Initial Submitters: Event Owner. Click Save.

## Step C – Actions

| Stage              | Action Type  | Configuration                                                                                                                                          |
|--------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Initial Submission | Email Alert  | Alert – Event Cancellation Request; template: Event Cancellation Request Notification; recipient: Role – Event Coordinator; From: Current User’s Email |
| Final Approval     | Field Update | Name: Event Status Update; Field: Event Status; New Value: Canceled                                                                                    |
| Final Approval     | Email Alert  | Alert – Cancellation Approved; template: Approval Notification to Event Owner; recipient: Event Owner                                                  |
| Final Rejection    | Email Alert  | Alert – Cancellation Rejected; template: Rejection Notification to Event Owner; recipient: Event Owner                                                 |
| Final Rejection    | Field Update | Name: Event Status To Rejected; Field: Event Status; New Value: Rejected                                                                               |

## Step D – Approval Step and Activation

1. Click Add New Step; Step Name: Manager Review; All records should enter this step; click Next.
2. Assign the approver automatically (a specific user / related user) and click Save.
3. Open View Approval Process Detail Page and click Activate.

The screenshot displays the Salesforce Classic Email Templates interface. The main heading is 'Classic Email Templates'. Below it, the 'Event Cancellation Request Notification' template is selected. The 'Email Template Detail' section shows the following information:

- Email Template Name: Event Cancellation Request Notification
- Template Unique Name: Event\_Cancellation\_Request\_Notification
- Encoding: Unicode (UTF-8)
- Author: Pratiksha S (Change)
- Description: Pratiksha S (Change)
- Created By: Pratiksha S, 9/30/2026, 11:47 PM
- Modified By: Pratiksha S, 10/1/2026, 12:01 AM

The 'Email Template' section shows the subject line: 'Approval Request: Cancel Event (Event\_\_cName)'. The 'Plain Text Preview' section shows the following text:

```

Hello!
An approval request has been submitted to cancel the event:
- Event Name: {Event__cName}
- Event Date: {Event__cEvent_Date__c}
Thank you,
EventForce Management System
  
```

## 6.2.6 Milestone 7 – Flow: 3-Day Reminder Email to Client

Scenario: when an Event is created or updated with Event Status = Confirmed and has a client with a valid email address, the system automatically sends a reminder email to the client three

days before the Event Date.

## Flow Configuration

1. Setup → Flows → New Flow → Record-Triggered Flow → Create.
2. Object: Event; Trigger: A record is created or updated.
3. Condition: Event Status equals Confirmed. Optimize the flow for Actions and Related Records; click Done.
4. Click Start → Add Schedule Path. Path Name: 3\_Days\_Before; Time Source: Event\_\_c: Event Date; Offset: 3 Days Before.
5. Toolbox → New Resource → Text Template. API Name: Subject; Body: “Reminder: Your Event {!\$Record.Name} is in 3 days” (plain text).
6. Create a second Text Template with API Name Body (content shown below).
7. Under the 3\_Days\_Before path click + → Action → Send Email. Label: Alert\_Client\_3Day\_Reminder.
8. Recipient Address List: Client Email; CC Recipient Address List: Event Owner.
9. Configure Email Content → Compose Email Content: Subject = Subject resource; Body = Body resource.
10. Click Save, enter Flow Label “Client Reminder - 3 Days Before”, then Save & Activate.

## Body Text Template

```
Hello {!$Record.Client__r.Name},

This is a reminder that your event "{!$Record.Name}" will take place on
{!$Record.Event_Date__c}.

Venue: {!$Record.Venue__r.Name}
Type: {!$Record.Event_Type__c}

We look forward to seeing you!

- EventForce Team
```

Figure 6.1 – Record-triggered flow

## 6.2.7 Milestone 8 – Apex Class & Trigger: Venue Status Update

Scenario: when an Event is created or updated with a Venue, the venue’s Availability Status is set to Reserved when the event is Confirmed, and set back to Available when the event is Canceled. The logic is kept in a helper class and invoked from a thin trigger – a recommended best practice.

### Apex Class: VenueStatusHelper

Open Gear Icon → Developer Console → File → New → Apex Class → name it VenueStatusHelper, then enter and save (Ctrl+S):

```
public with sharing class VenueStatusHelper {
    public static void updateVenueStatus(List<Event__c> events){
        List<Venue__c> venues = new List<Venue__c>();
        for(Event__c ev : events){
            if(ev.Venue__c != null){
                if(ev.Event_Status__c == 'Confirmed'){
```



```

        venues.add(new Venue_c(Id=ev.Venue_c,
Availability_Status_c='Reserved'));
    } else if(ev.Event_Status_c == 'Canceled'){
        venues.add(new Venue_c(Id=ev.Venue_c,
Availability_Status_c='Available'));
    }
}
if(!venues.isEmpty()) update venues;
}
}

```

### Apex Trigger: EventTrigger13

File → New → Apex Trigger → Name: EventTrigger13, sObject: Event\_\_c:

```

trigger EventTrigger13 on Event__c (after insert, after update) {
    VenueStatusHelper.updateVenueStatus(Trigger.new);
}

```

The screenshot shows the Salesforce Developer Console interface. The top pane displays the Apex Trigger 'EventTrigger13' on the 'Event\_\_c' object, configured for 'after insert, after update' events. The trigger code is as follows:

```

1 trigger EventTrigger13 on Event__c (after insert, after update) {
2     if (Trigger.isAfter) {
3         VenueStatusHelper.updateVenueStatus(Trigger.new, Trigger.oldMap);
4     }
5 }

```

The bottom pane shows the 'Open' dialog box, which is used to open a file or package. The dialog has two main sections: 'Entities' and 'Related'. The 'Entities' section lists various entities, including 'VenueStatusHelper', which is highlighted. The 'Related' section shows the relationships between the selected entity and other entities in the system.

| Entity Type      | Entities                  | Related    |
|------------------|---------------------------|------------|
| Entity Type      | Name                      | Namespace  |
| Classes          | DeveloperEditionUtils     | devedapp   |
| Triggers         | DeveloperEditionUtilsT... | devedapp   |
| Pages            | PostInstallScript         | devedapp   |
| Page Components  | PostInstallScriptTest     | devedapp   |
| Objects          | VenueStatusHelper         | ← Event__c |
| Static Resources | BatchCompleteEvents       | ← Event__c |
| Packages         | EventForceTest            | ← Venue__c |
|                  | ScheduleCompleteEve...    | ← Venue__c |

The 'Open' dialog box also includes a 'Filter' field and a 'Refresh' button. The bottom of the console shows the 'Logs' tab, which displays a table with columns: User, Application, Operation, Time, Status, Read, and Size.

### 6.2.8 Milestone 9 – Apex Trigger: Prevent Double Booking

Scenario: two events must not be allowed at the same Venue on the same date. The before-insert/before-update trigger collects the venues involved, queries existing events at those venues in a single query (bulk-safe), and raises an error if another event already exists on the same date.

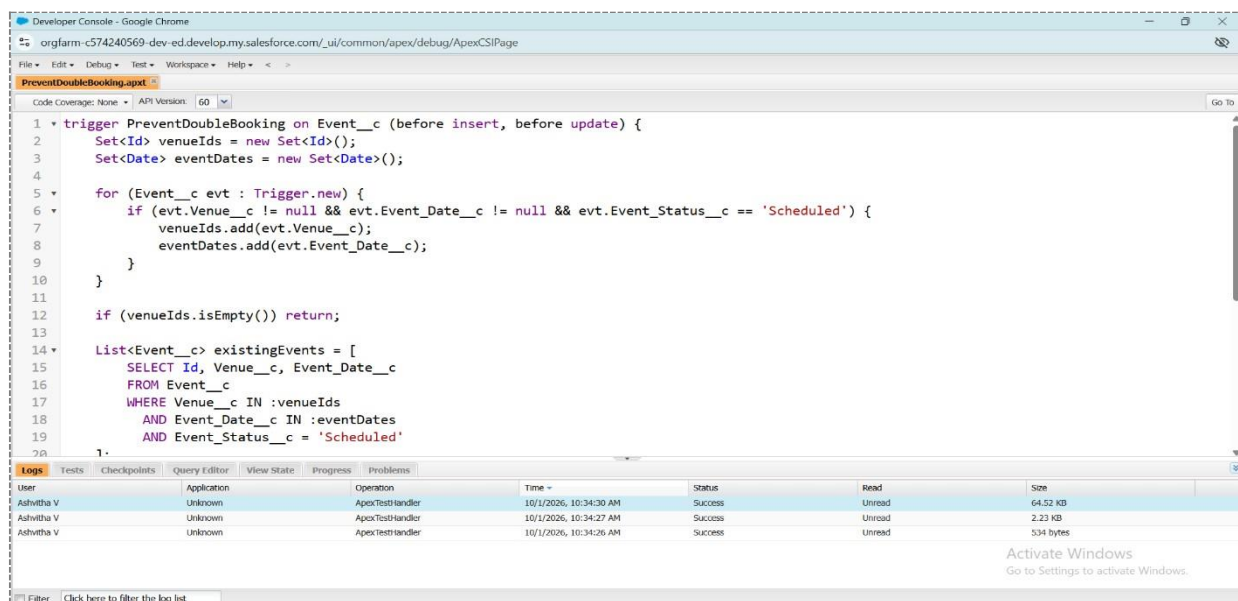
```
trigger PreventDoubleBooking on Event__c (before insert, before update) {
    Set<Id> venueIds = new Set<Id>();
    for(Event__c ev : Trigger.new){
        if(ev.Venue__c != null) venueIds.add(ev.Venue__c);
    }

    Map<Id, List<Event__c>> venueEventMap = new Map<Id, List<Event__c>>();
    for(Event__c e : [SELECT Id, Venue__c, Event_Date__c
                      FROM Event__c WHERE Venue__c IN :venueIds]){
        if(!venueEventMap.containsKey(e.Venue__c)) {
```

```

        venueEventMap.put(e.Venue__c, new List<Event__c>());
    }
    venueEventMap.get(e.Venue__c).add(e);
}

for(Event__c ev : Trigger.new){
    if(ev.Venue__c != null && venueEventMap.containsKey(ev.Venue__c)){
        for(Event__c existing : venueEventMap.get(ev.Venue__c)){
            if(existing.Event_Date__c == ev.Event_Date__c &&
existing.Id != ev.Id){
                ev.addError('This Venue is already booked on this date.');
```



## 6.2.9 Milestone 10 – Batch & Schedulable Apex

Scenario: mark all past events as Completed every night. A Batch Apex class processes the records in chunks, and a Schedulable class runs the batch automatically.

### Batch Class: BatchCompleteEvents

```

global class BatchCompleteEvents implements Database.Batchable<sObject> {
    global Database.QueryLocator start(Database.BatchableContext bc){
        return Database.getQueryLocator(
            'SELECT Id, Event_Date__c, Event_Status__c FROM Event__c ' +
            'WHERE Event_Date__c < TODAY AND Event_Status__c != \'Completed\''
        );
    }
    global void execute(Database.BatchableContext bc, List<Event__c> scope){
        for(Event__c ev : scope){
            ev.Event_Status__c = 'Completed';
        }
        update scope;
    }
    global void finish(Database.BatchableContext bc){
        System.debug('Past events marked as Completed');
```

**Schedulable Class: ScheduleCompleteEvents**

```
global class ScheduleCompleteEvents implements Schedulable {
```

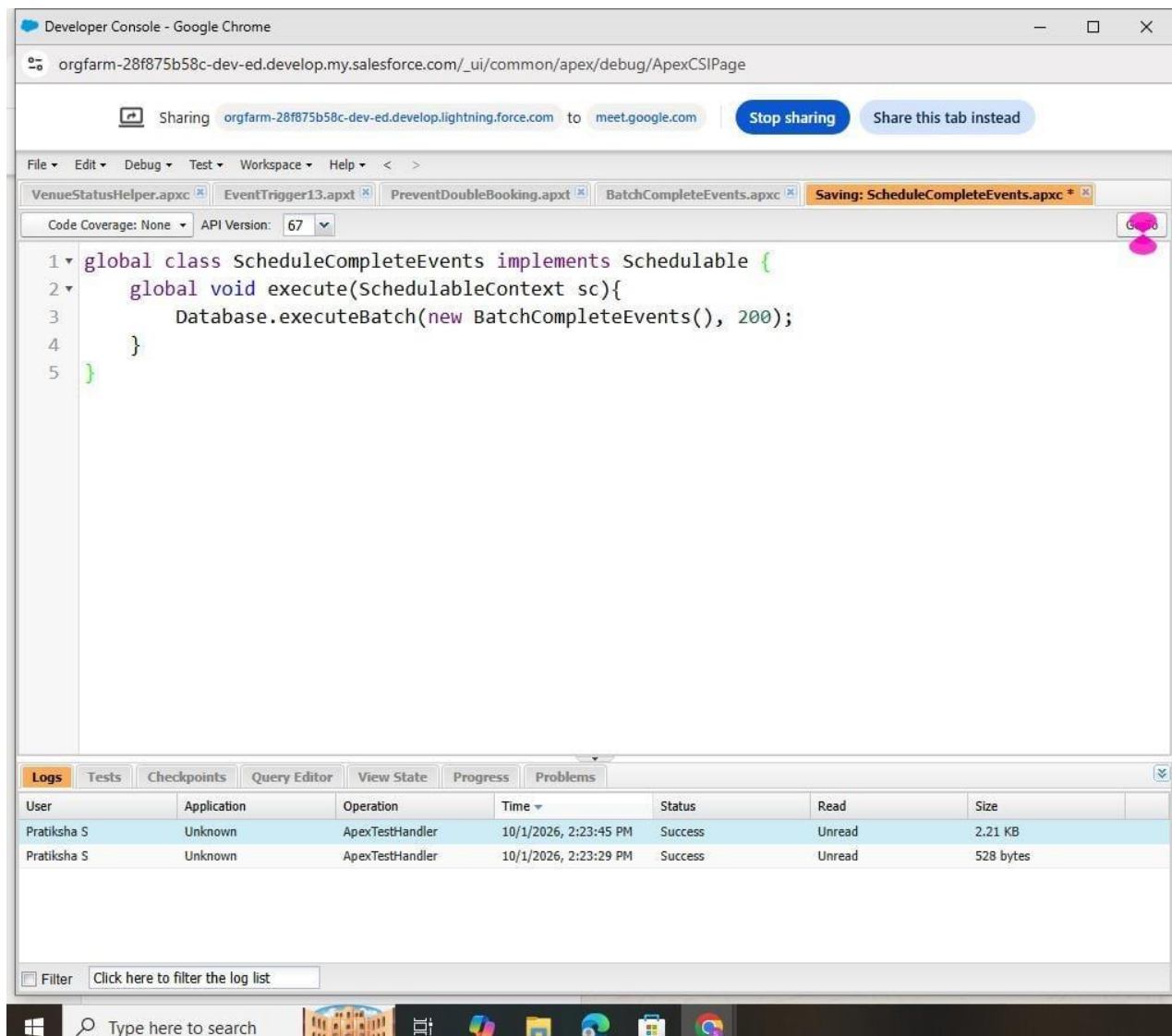
```

global void execute(SchedulableContext sc){
    Database.executeBatch(new BatchCompleteEvents(), 200);
}
}

```

### Scheduling the Job

1. Setup → Apex Classes → click Schedule Apex.
2. Job Name: Daily Event Completion; Apex Class: ScheduleCompleteEvents.
3. Frequency: Weekly; Days: select all days; Preferred Start Time: 08:00 PM.
4. Click Save. The batch now runs every night at 8 PM and marks past events as Completed.



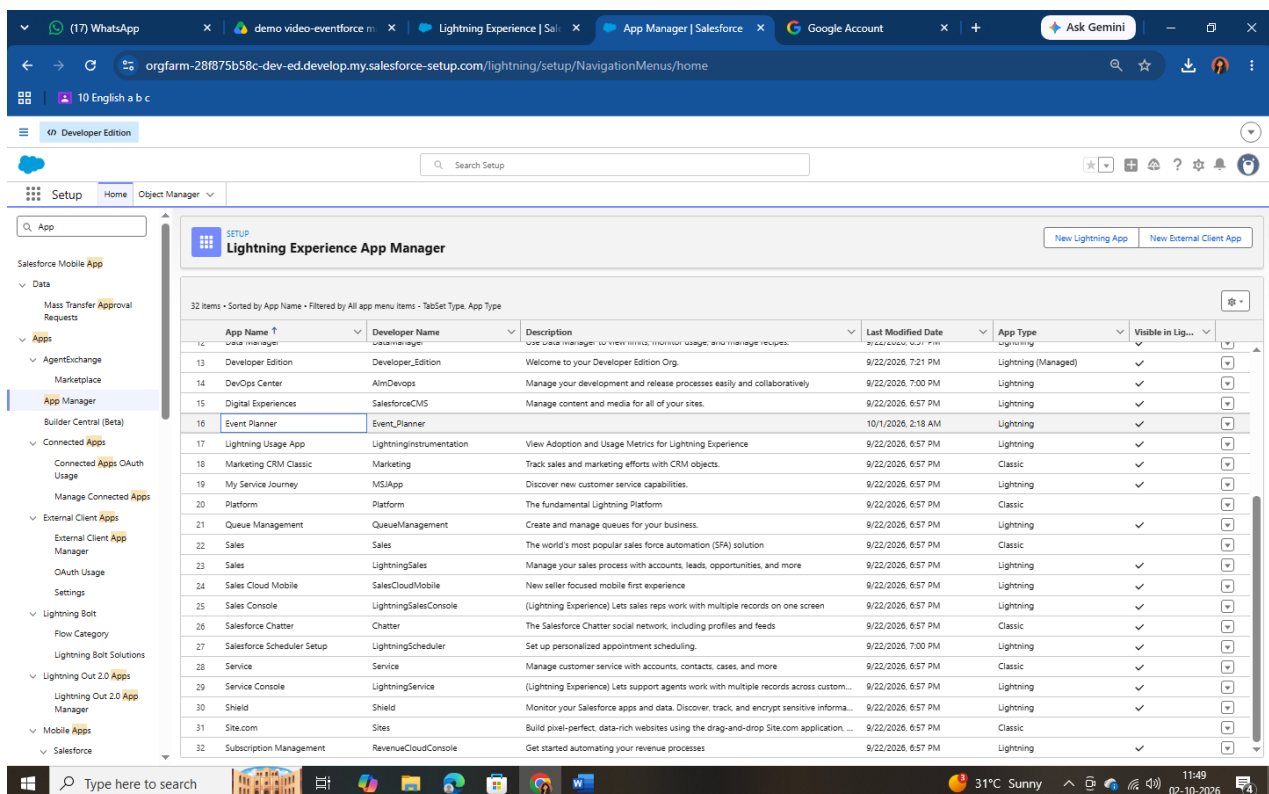
## 6.3 Phase 3: UI/UX Development & Customization

This phase created a simple, user-friendly interface: a custom Lightning App that groups all modules into one workspace, and reports and dashboards that visualize event schedules, vendor performance and client feedback.

### 6.3.1 Milestone 11 – The Lightning App

An app is a group of tabs that work together as a unit, with a name, a logo and a set of tabs. The Event Planner app was created as follows:

1. Setup → App Manager → New Lightning App. App Name: Event Planner; upload an image; click Next.
2. Leave App Options and Utility Items at their defaults and click Next.
3. From Available Items select Event, Client, Vendor, Venue, Feedback, Reports and Dashboards and move them to Selected Items; click Next.
4. From Available Profiles select System Administrator (and other required profiles) and move them to Selected Profiles.
5. Click Save & Finish. Verify by opening the App Launcher, searching Event Planner and selecting the app.



### 6.3.2 Milestone 12 – Reports

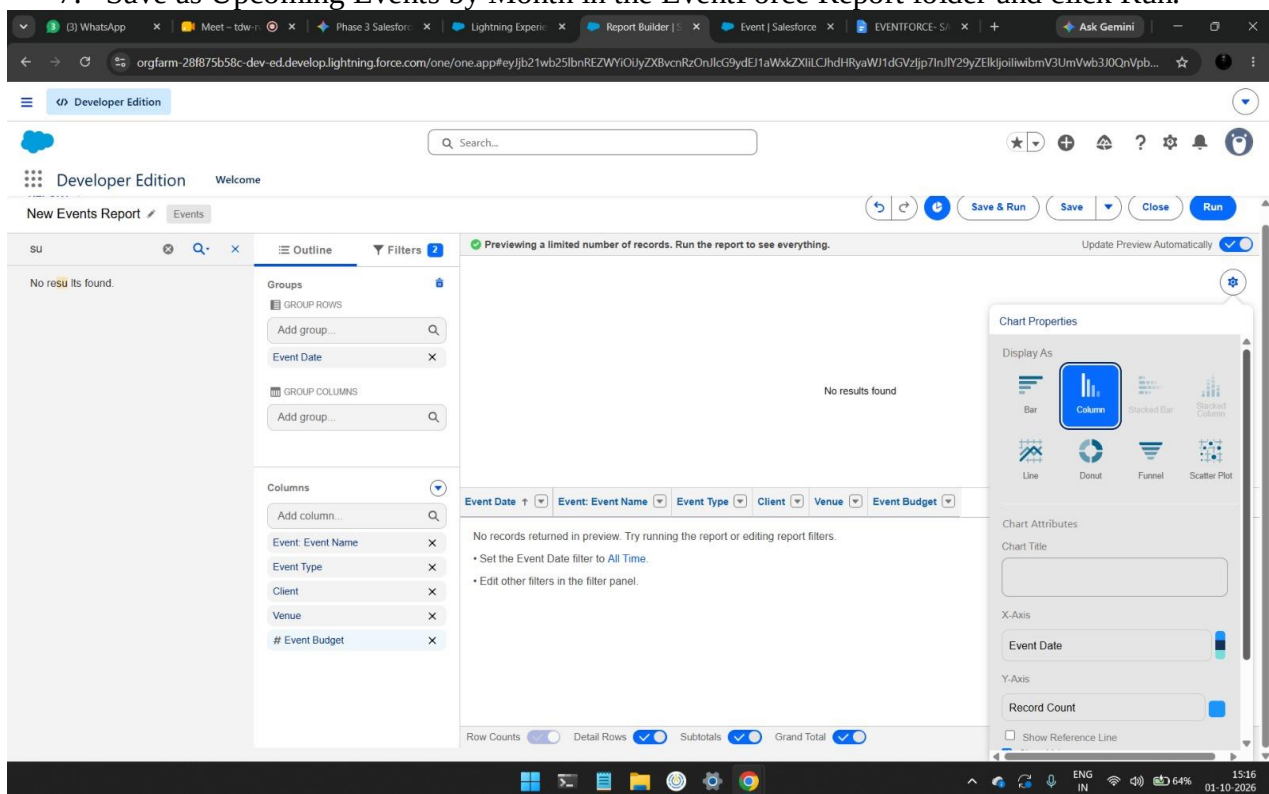
#### Creating and Sharing the Report Folder

1. App Launcher → Reports → New Folder. Folder Label: EventForce Report; click Save.
2. From All Folders, open the drop-down on the folder → Share.
3. Share with Roles: Event Coordinator and Vendor Manager with View access; click Share, then Done.

#### Report: Upcoming Events by Month

Purpose: view upcoming events and their counts and budgets by month. At least 10 records were created in every object before building the report.

1. Reports → New Report; category All; select Events (or Events with Client & Venue); Start Report.
2. Columns: Event Name, Event Date, Event Type, Client Name, Venue Name, Event Budget.
3. Filters: Show Me = All Events; Event Date  $\geq$  TODAY (or Next 30 Days); Event Status  $\neq$  Canceled.
4. Group Rows: Event Date → Group by Calendar Month.
5. Add Summary: Row Count and SUM of Event Budget.
6. Add Chart: Column or Line; X-axis = Event Date (Calendar Month); Y-axis = Row Count or SUM(Event Budget).
7. Save as Upcoming Events by Month in the EventForce Report folder and click Run.



The image displays two screenshots of the Salesforce Developer Edition interface, specifically the Report Builder for a report named 'New Events Report'.

**Top Screenshot:** The report is in 'Preview' mode, showing 'No records returned in preview. Try running the report or editing report filters.' The left sidebar shows the report structure with 'Event Date' as the group and 'Event Budget' as the column. A dropdown menu is open for the 'Event Budget' column, showing aggregation options: Sum (selected), Average, Max, Min, and Median. The bottom of the interface shows row counts and detail rows.

**Bottom Screenshot:** The report is in 'Preview' mode, showing 'No results found'. The left sidebar shows the report structure with 'Event Date' as the group and 'Event Budget' as the column. The 'Chart Properties' panel is open on the right, showing options for chart type (Line, Donut, Funnel, Scatter Plot) and attributes: Chart Title, X-Axis (Event Date), and Y-Axis (Record Count). The bottom of the interface shows row counts and detail rows.



**Save Report**

**Report Name**  
Upcoming Events by Month

**Report Unique Name**  
Upcoming\_Events\_by\_Month\_Ufo

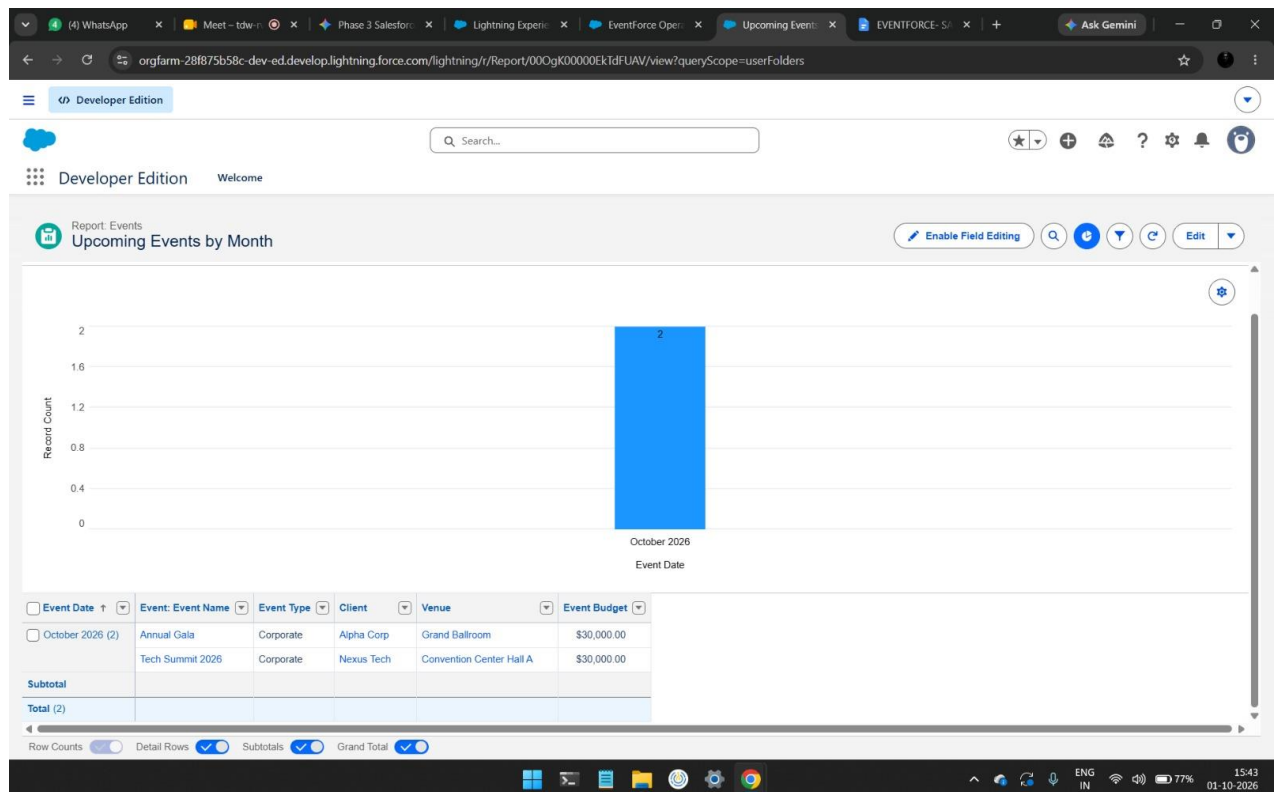
**Report Description**

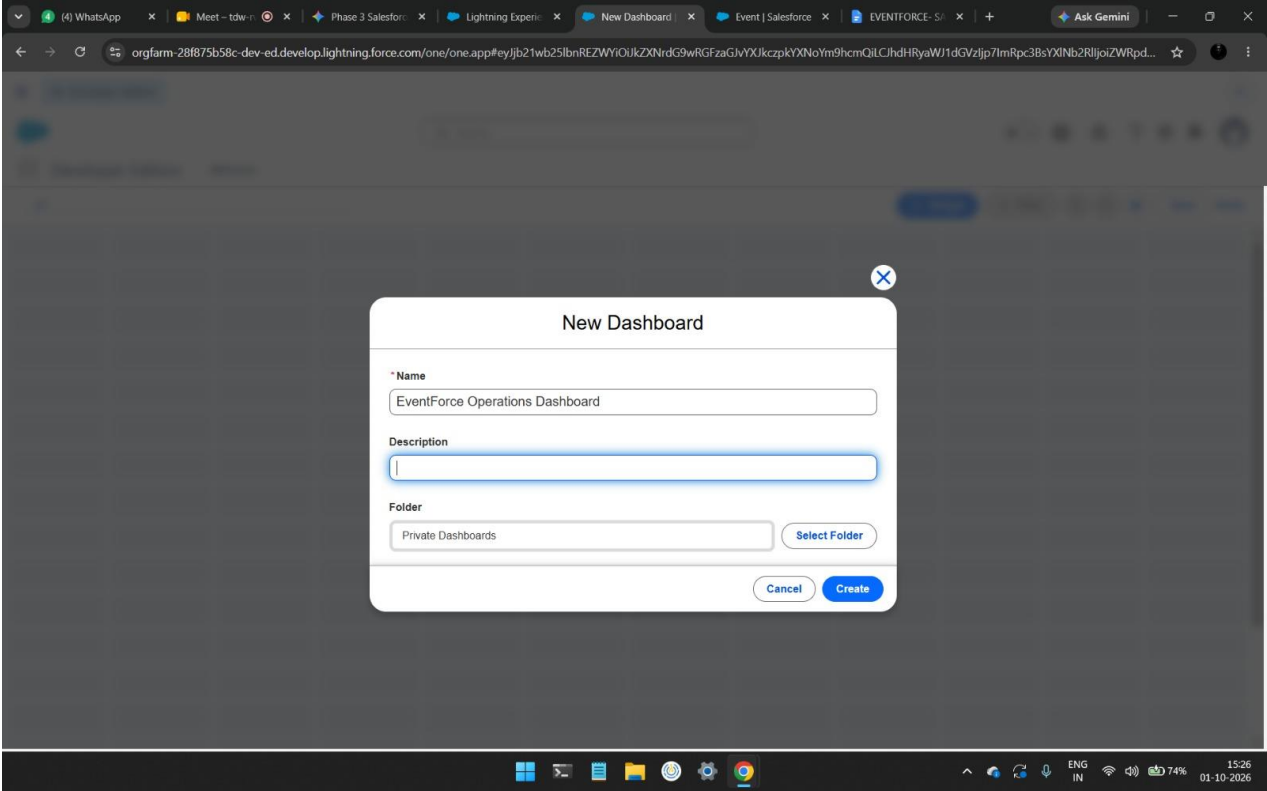
**Folder**  
Private Reports [Select Folder](#)

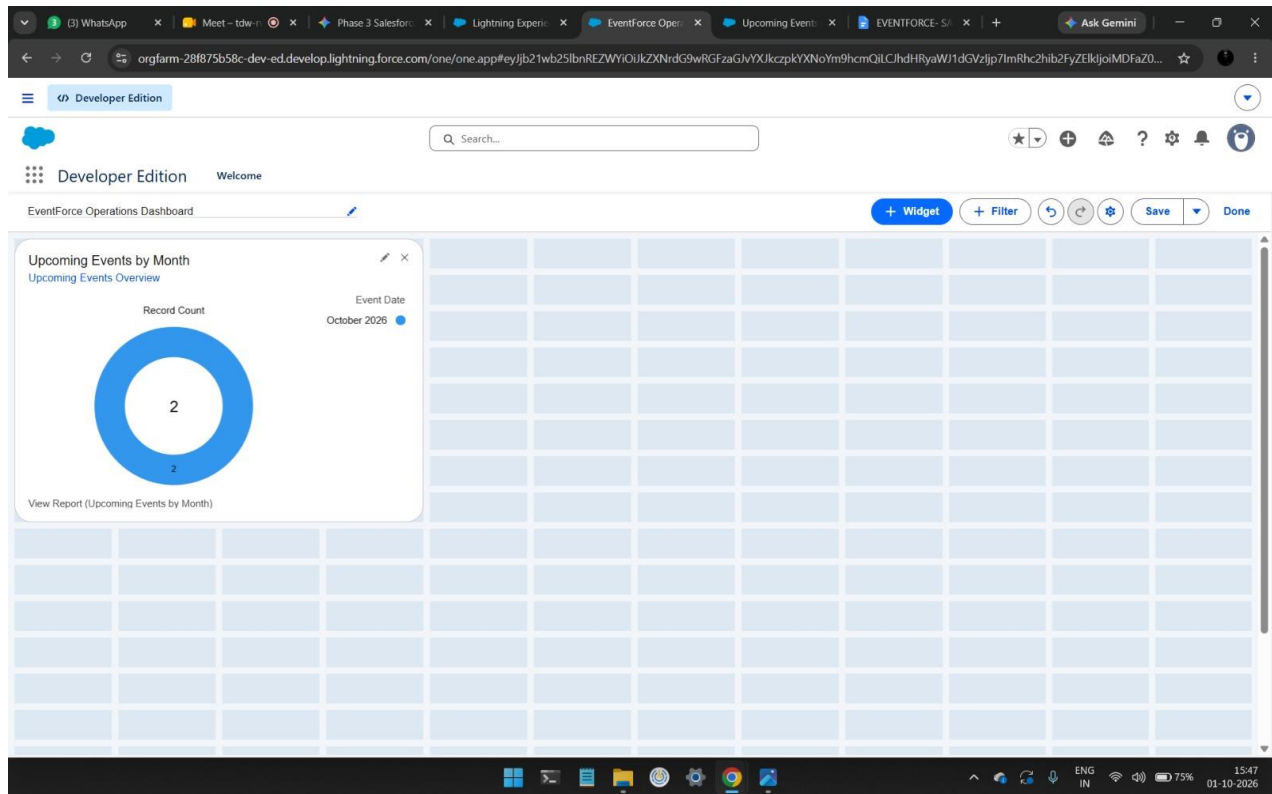
[Cancel](#) [Save](#)

### 6.3.3 Milestone 13 – Dashboards

1. Open the Dashboards tab in the Event Planner app → New Dashboard.
2. Name: EventForce Operations Dashboard; click Create.
3. Click + Widget and select the Upcoming Events by Month report.
4. Chart: Donut; Max Values Displayed: 6; Subtitle: Upcoming Events Overview; Widget Theme: Light.
5. Click Save.







## 6.4 Phase 4: Data Migration, Testing & Security

This phase prepared the system for use by loading validated data, creating users and implementing role-based access control. Records for Events, Clients, Vendors, Venues and Feedback were imported with the Data Import Wizard after pre-validation checks to avoid duplicates. Four types of user accounts (Event Admin, Event Coordinator, Vendor Manager and Client) were created with profiles and roles aligned to their access needs. Organization-Wide Defaults were set to Private and extended by sharing rules.

### 6.4.1 Milestone 14 – Profiles

A profile is a collection of settings and permissions that define what a user can do in Salesforce (object and field permissions, tab and app settings, login hours, etc.). A user has exactly one profile, but a profile can be assigned to many users.

#### Event Admin Profile

Created by cloning System Administrator (Setup → Profiles → New Profile → Existing Profile: System Administrator → Profile Name: Event Admin). Cloning provides Modify All Data, Customize Application and full CRUD access on all objects, which suits the administrator role.

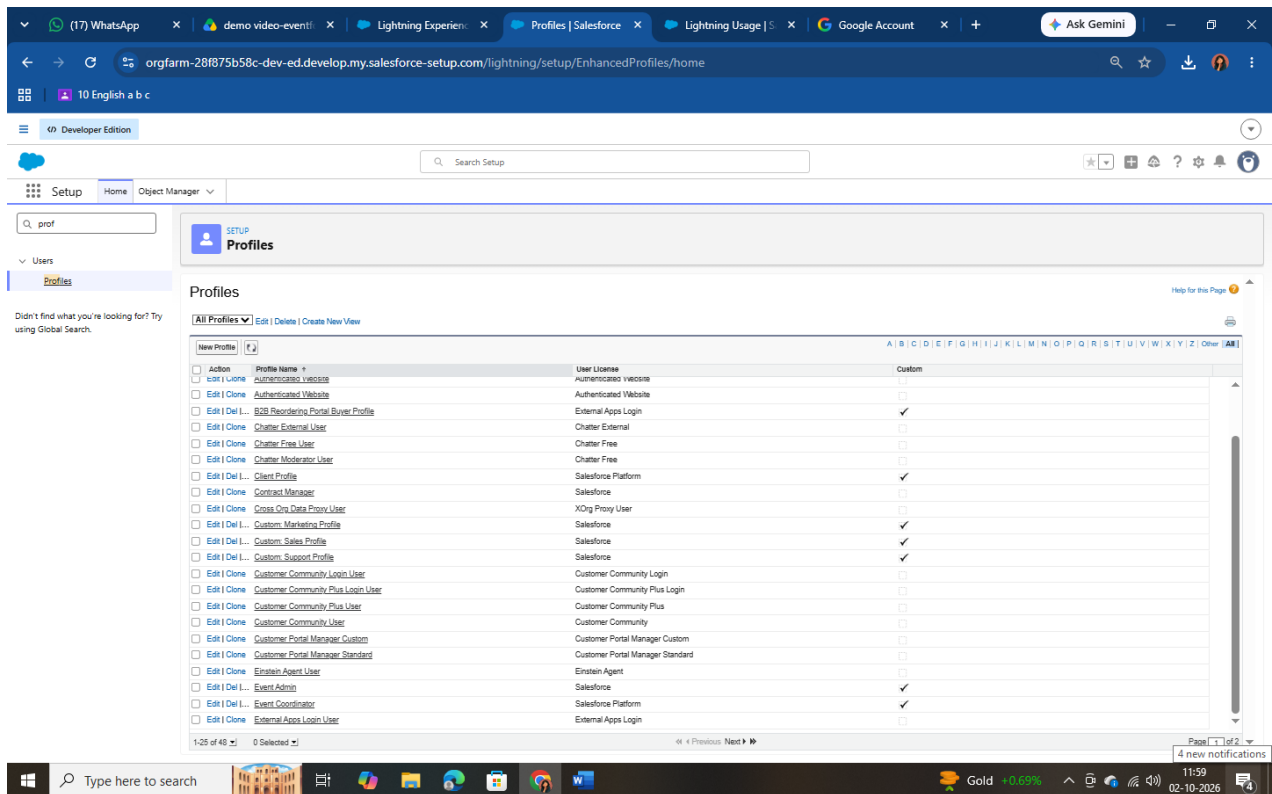
#### Custom Profiles for Other Users

The Event Coordinator, Vendor Manager and Client Profile were each cloned from Standard Platform User, then edited: Custom App Settings → default app set to Event Planner; Custom

Object Permissions set as below; session timeout set to 2 hours of inactivity; password never expires; minimum password length 8.

| Object   | Event Coordinator          | Vendor Manager             | Client Profile         |
|----------|----------------------------|----------------------------|------------------------|
| Event    | Create, Read, Edit, Delete | Read only                  | Read only (own events) |
| Client   | Read, Edit                 | Read only                  | —                      |
| Vendor   | Read                       | Create, Read, Edit, Delete | No access              |
| Venue    | Read                       | Read only                  | No access              |
| Feedback | Create, Read               | Read only                  | Create, Read           |

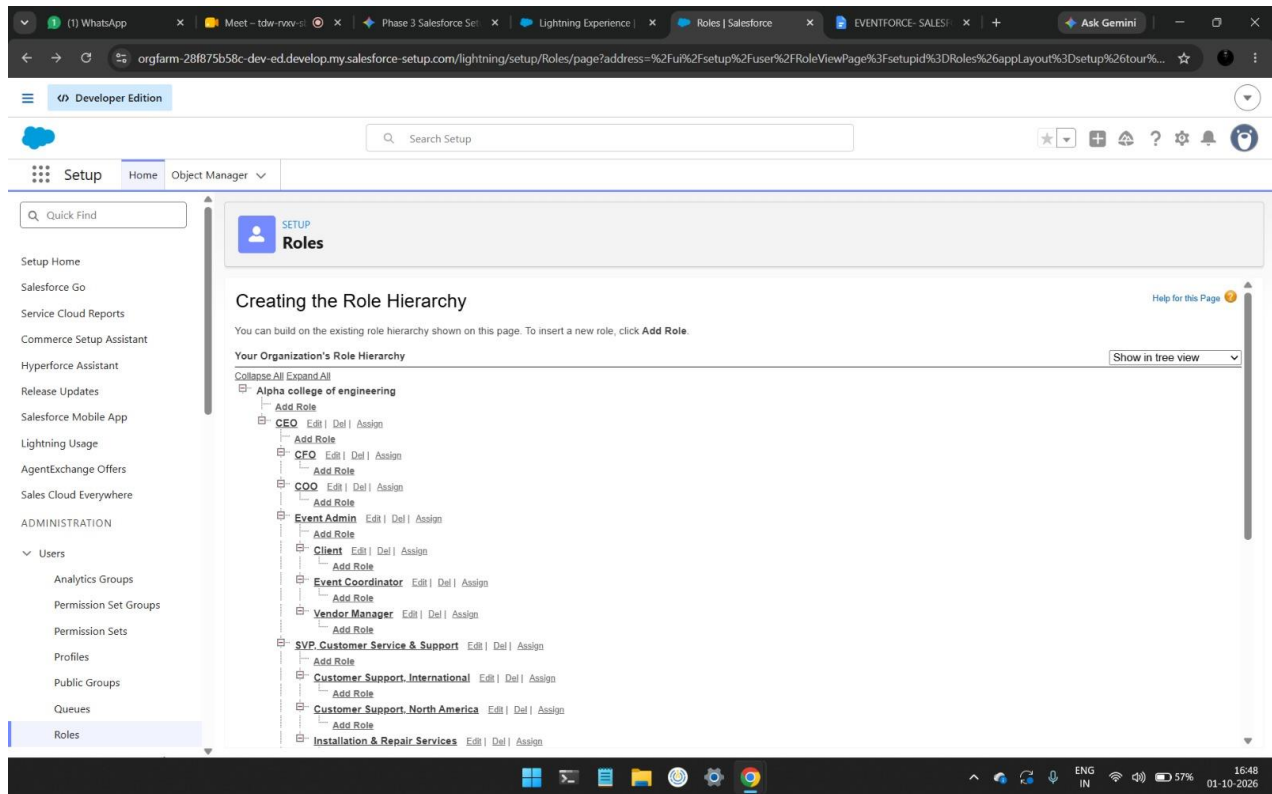
The screenshot shows the Salesforce Setup interface for the 'Profiles' page. The page title is 'Profiles' and it includes a search bar and a 'Search Setup' button. The left sidebar shows the 'Setup' menu with 'Profiles' selected. The main content area displays a table of profiles with the following columns: Action, Profile Name, User License, and Custom. The table lists various profiles, including 'Analytics Cloud Integration User', 'Authenticated Website', 'B2B Reordering Portal Buyer Profile', 'Chatter External User', 'Chatter Free User', 'Chatter Moderator User', 'Client Profile', 'Contract Manager', 'Cross Org Data Proxy User', 'Custom Marketing Profile', 'Custom Sales Profile', 'Custom Support Profile', 'Customer Community Login User', 'Customer Community Plus Login User', 'Customer Community Plus User', 'Customer Community User', 'Customer Portal Manager Custom', 'Customer Portal Manager Standard', and 'Einstein Agent User'. The 'Custom' column shows a checkmark for the 'Salesforce' profile. The bottom of the page shows the Windows taskbar with the search bar and system tray.



### 6.4.2 Milestone 15 – Roles

Roles define the reporting hierarchy and determine record visibility upward through the hierarchy. The Event Admin role was created under CEO (Setup → Roles → Set Up Roles → Expand All → Add Role), and three roles were added beneath it.

| Role              | Reports To  |
|-------------------|-------------|
| Event Admin       | CEO         |
| Event Coordinator | Event Admin |
| Vendor Manager    | Event Admin |
| Client            | Event Admin |



### 6.4.3 Milestone 16 – Users

A user is anyone who logs in to Salesforce. Users were created from Setup → Users → New User. The first user was created with Role Event Admin, License Salesforce and Profile Event Admin. The other users were created with License Salesforce Platform and the matching Role and Profile.

| User Type         | Role              | User License        | Profile           |
|-------------------|-------------------|---------------------|-------------------|
| Event Admin       | Event Admin       | Salesforce          | Event Admin       |
| Event Coordinator | Event Coordinator | Salesforce Platform | Event Coordinator |
| Vendor Manager    | Vendor Manager    | Salesforce Platform | Vendor Manager    |
| Client            | Client            | Salesforce Platform | Client Profile    |

Note: a Developer Edition org has a limited number of Salesforce licenses. If none are available, deactivate an existing user holding that license or change the license type.

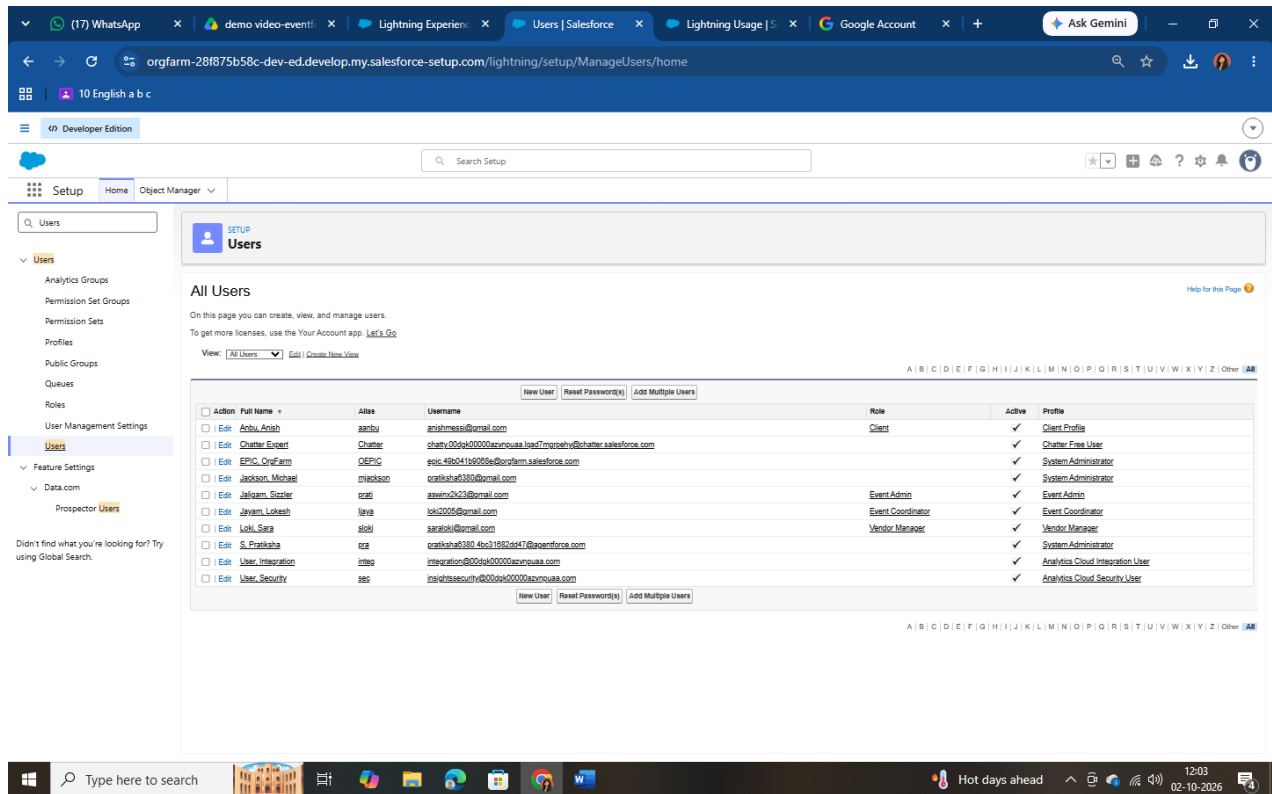
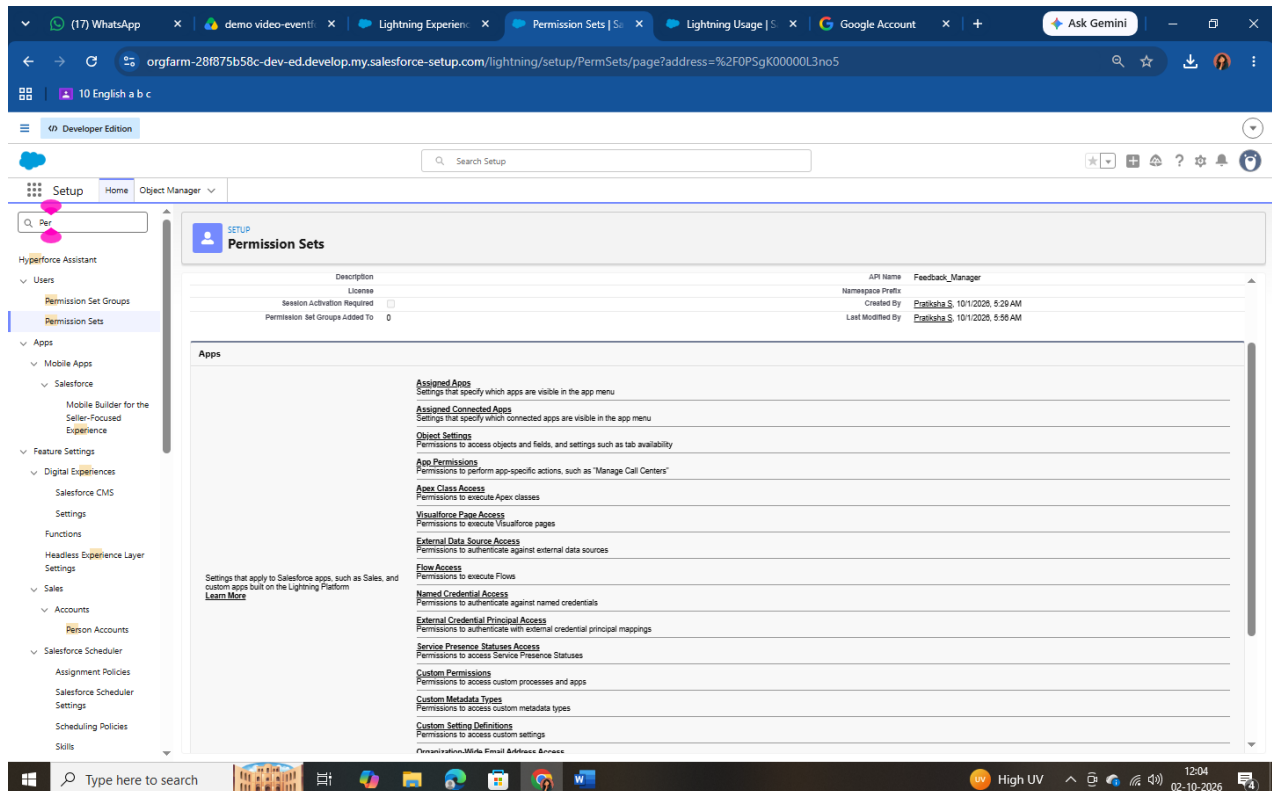


Figure 6.2 – Users

#### 6.4.4 Milestone 17 – Permission Set

A permission set grants extra permissions without changing a user's profile. A Feedback Manager permission set was created to give selected users full access to the Feedback object.

1. Setup → Permission Sets → New. Label: Feedback Manager; click Save.
2. Open Object Settings → Feedback → Edit; enable Read, Create, Edit and Delete; click Save.
3. Click Manage Assignments → Add Assignment, select the Event Coordinator user, and click Next → Assign.



### 6.4.5 Milestone 18 – Sharing Settings

Organization-Wide Defaults (OWD) set the baseline access to records, which can then be opened up using the role hierarchy, sharing rules, manual sharing or Apex sharing. Sharing rules can only extend access, never restrict it.

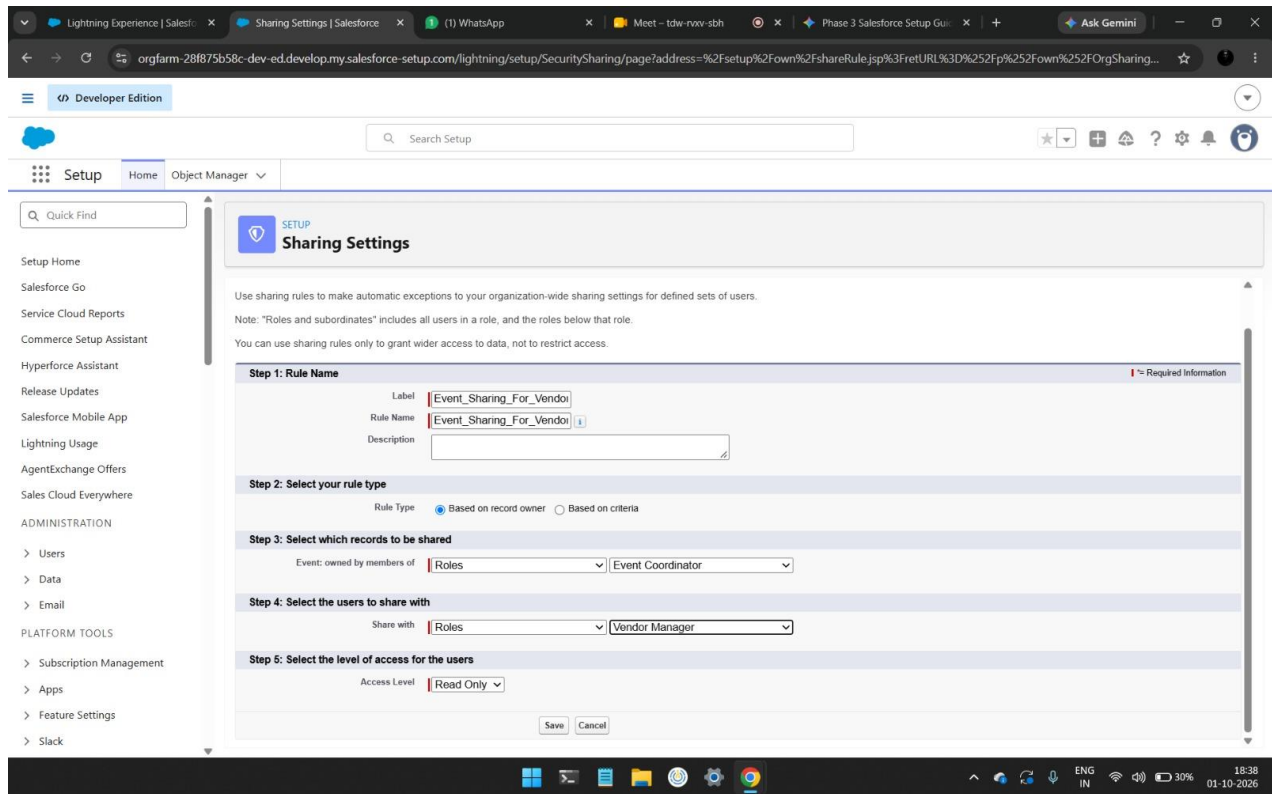
#### Configuring OWD

1. Setup → Sharing Settings → Edit (Organization-Wide Defaults).
2. For the Event object set Default Internal Access = Private and Default External Access = Private.
3. Click Save.

#### Sharing Rule: Event\_Sharing\_For\_Vendors

1. In Sharing Settings find the Event object's Sharing Rules and click New.
2. Label: Event\_Sharing\_For\_Vendors (rule name auto-populates).
3. Records to be shared: members of Roles → Event Coordinator.
4. Share with: Roles → Vendor Manager.
5. Access Level: Read Only; click Save.





## 6.5 Phase 5: Deployment, Documentation & Maintenance

### 6.5.1 Deployment Strategy

All configurations, customizations and automations were built and tested within a Salesforce Developer Edition Org. The objective of this phase is to understand how a real-world Salesforce CRM solution is prepared for deployment after configuration, testing and security setup. Actual production deployment is not performed in this project because Developer Edition orgs are standalone and not connected to production instances, and real deployments require sandboxes, change sets or DevOps tooling.

In an enterprise setting, the recommended deployment path would be:

1. Develop and unit-test in a Developer or Developer Pro sandbox.
2. Promote the metadata (objects, fields, flows, Apex, approval process, reports, dashboards) through a Partial/Full sandbox for UAT using Change Sets or Salesforce DX / Git-based CI/CD pipelines.
3. Run Apex tests (minimum 75% code coverage is required for production deployment) and validate the deployment.
4. Deploy to production during a low-usage window, activate flows, approval processes and scheduled jobs, and assign permission sets.
5. Load production data, train users and provide post-go-live support.

**Note:** Apex test classes were not part of this academic build. Before any production deployment, test classes for VenueStatusHelper, PreventDoubleBooking, BatchCompleteEvents and ScheduleCompleteEvents must be written to meet the 75% coverage requirement.

### 6.5.2 Maintenance, Monitoring & Troubleshooting

Maintenance keeps event, client, vendor and feedback data accurate and the automations healthy.

| Activity      | Approach                                                                                                            |
|---------------|---------------------------------------------------------------------------------------------------------------------|
| Monitoring    | Regularly review Flow error emails, Apex Jobs and Scheduled Jobs, and pending approvals                             |
| Access issues | Resolve user-access errors and permission conflicts by reviewing profiles, permission sets, roles and sharing rules |
| Debugging     | Use Debug Logs and the Developer Console to debug Apex triggers and Flow logic                                      |
| Rule tuning   | Adjust validation rules and sharing settings as business needs evolve                                               |
| Data quality  | Run duplicate checks and periodic data cleansing; review reports for anomalies                                      |
| Releases      | Review Salesforce seasonal release notes and re-test critical automations after each release                        |

### Common Issues and Resolutions

| Issue                                          | Likely Cause                                               | Resolution                                                                  |
|------------------------------------------------|------------------------------------------------------------|-----------------------------------------------------------------------------|
| User cannot see the Event Planner app or a tab | Profile or permission set lacks app/tab access             | Enable the app/tab visibility in the profile or permission set              |
| Approval process cannot be submitted           | No approver/manager configured, or entry criteria not met  | Assign a manager on the User record; confirm status is Pending Cancellation |
| Reminder email not sent                        | Flow inactive, status not Confirmed, or client email blank | Activate the flow; verify the status and client email                       |
| Save blocked on an event                       | Double-booking trigger fired                               | Choose a different date or venue                                            |
| Unable to create a user                        | No free Salesforce license in the Developer Org            | Deactivate an unused user or change the license type                        |

### 6.5.3 Documentation Summary

This document records the purpose, data model, configuration, automation, security and testing of the EventForce Management System. It covers:

- Business overview and objectives.
- Phase-wise implementation: Requirement → Development → UI → Testing → Security.
- Data model and relationships (Event, Client, Vendor, Venue, Feedback, Event Vendor).
- Automation components: Flow, Approval Process, Validation Rule and Apex logic.
- Security model: Profiles, Roles, Permission Sets, OWD and Sharing Rules.
- Testing results and screenshots of all Salesforce features.

It provides a blueprint for future developers and administrators, supporting consistent development, training and smooth knowledge transfer.

## 7. FUNCTIONAL & PERFORMANCE TESTING

During testing, screenshots were captured from Salesforce to validate event creation, venue assignment, double-booking prevention triggers, approval status updates, batch processing, scheduled reminders, and dashboard analytics. All features were verified to ensure correct functionality and performance.

Testing validated data accuracy, automation performance and user permissions. The main test scenarios are summarized below.

| #  | Area               | Test Scenario                                            | Expected Result                                             |
|----|--------------------|----------------------------------------------------------|-------------------------------------------------------------|
| 1  | Validation Rule    | Save a Client with an invalid email (e.g. “abc@”)        | Save blocked with “Please Enter Valid Email Address”        |
| 2  | Formula Field      | Create events of each Event Type                         | Event Budget auto-populates as per formula                  |
| 3  | Double Booking     | Create two events at the same Venue on the same date     | Error: “This Venue is already booked on this date.”         |
| 4  | Venue Status       | Set an event to Confirmed, then Canceled                 | Venue becomes Reserved, then Available                      |
| 5  | Approval Process   | Set Event Status to Pending Cancellation and submit      | Approval request and alert emails are sent                  |
| 6  | Approval – Approve | Approver approves the request                            | Status = Canceled; owner notified                           |
| 7  | Approval – Reject  | Approver rejects the request                             | Status = Rejected; owner notified                           |
| 8  | Flow               | Confirm an event dated 3+ days ahead with a client email | Reminder email is sent three days before the event date     |
| 9  | Batch Apex         | Run the batch with past-dated events                     | Past events are marked Completed                            |
| 10 | Lookup Filter      | Select a Client on Feedback, then open the Event lookup  | Only that client’s events are listed                        |
| 11 | Security           | Log in as each user type                                 | Each user sees only the tabs, objects and records permitted |
| 12 | Sharing Rule       | Vendor Manager views coordinator-owned events            | Events are visible as Read Only                             |

## 8. ADVANTAGES & DISADVANTAGES

### Advantages

- **Centralized Data Management** – Event, Client, Vendor, Venue, Feedback, and EventVendor information is managed within Salesforce.
- **Process Automation** – Flows automate reminders and other business processes, reducing manual effort.
- **Prevents Double Booking** – Apex Trigger and VenueStatusHelper validate venue availability and help prevent booking conflicts.
- **Controlled Cancellation Process** – Approval Process provides structured handling of event cancellation requests.
- **Improved Operational Visibility** – Reports and Dashboards provide visibility into event schedules, vendor activities, and feedback.
- **Secure Access** – Profiles, Roles, Permission Sets, OWD, and Sharing Rules provide controlled access to records.
- **Scalable Solution** – Salesforce provides a platform that can support future growth and additional business requirements.
- **Better Data Quality** – Validation Rules and pre-validation during data migration help maintain accurate and consistent data.

### Disadvantages

- **Developer Edition Limitation** – The project was implemented in a Salesforce Developer Edition for academic purposes and was not connected to a production environment.
- **Production Deployment Not Performed** – Actual migration to production was outside the scope of this implementation.
- **Deployment Requires Additional Tools** – Real-world deployment may require sandboxes, Change Sets, Salesforce DX, or DevOps/CI-CD tools.
- **Configuration Complexity** – Managing Flows, Apex, Approval Processes, and security configurations requires appropriate Salesforce knowledge.
- **Ongoing Maintenance Required** – Flows, Apex jobs, approval processes, validation rules, and sharing settings need regular monitoring and troubleshooting.

## 9. CONCLUSION

The EventForce Management System successfully demonstrates how Salesforce can be used to streamline and automate event planning and management operations. The project brings Events, Clients, Vendors, Venues, Feedback, and EventVendor data into a centralized platform.

The implementation combines Salesforce Lightning, Flows, Validation Rules, Formula Fields, Approval Processes, Apex Triggers, Apex Classes, Batch Apex, Scheduled Apex, security controls, Reports, and Dashboards to address key business requirements. Automation supports activities such as event reminders, cancellation approvals, venue availability management, double-booking prevention, and event completion.

The project also demonstrates the importance of data migration, testing, security, documentation, and maintenance in a Salesforce implementation. Although the solution was developed and tested in a Developer Edition Org and actual production deployment was not performed, the project reflects real-world Salesforce implementation practices and provides a foundation for future scalability and enhancement.

### Key Learnings

- Designing a scalable data model, including junction objects for many-to-many relationships.
- Combining declarative tools (Flows, Approval Processes, Validation Rules) with Apex for advanced logic.
- Writing bulk-safe Apex triggers and using Batch and Schedulable Apex for asynchronous processing.
- Implementing layered security with Profiles, Roles, Permission Sets, OWD and Sharing Rules.
- Building reports and dashboards that turn operational data into decisions.
- Understanding how a Developer Org build maps to real-world deployment, maintenance and documentation practices.

## 10. FUTURE SCOPE

The EventForce Management System can be further enhanced to support larger-scale event operations and improve automation, integration, and user experience.

- **Production Deployment** – Deploy the solution from development to a production Salesforce environment using appropriate deployment tools.
- **Advanced Event Automation** – Extend automation for additional event planning and operational activities.
- **Enhanced Vendor Management** – Introduce more advanced vendor tracking and performance management capabilities.
- **Advanced Analytics** – Add more reports and dashboards for deeper insights into events, venues, vendors, and client feedback.
- **External Integrations** – Integrate Salesforce with external systems and services required for real-world event operations.
- **Mobile Accessibility** – Enhance the solution for convenient event management through mobile devices.
- **Scalability Enhancements** – Expand the system to support larger numbers of events, clients, vendors, venues, and users.
- **Continuous Maintenance & Optimization** – Regularly improve Flows, Apex, validation rules, sharing settings, and other configurations as business requirements evolve.

These enhancements can help transform the current EventForce implementation into a more comprehensive and enterprise-ready event management CRM solution.

## 11. APPENDIX

- **A. Custom Objects:** Event, Client, Vendor, Venue, EventVendor, Feedback
- **B. Automation:** Flows, Validation Rules, Approval Process, Apex Triggers & Classes
- **C. Processing:** Batch Apex and Scheduled Apex
- **D. Security:** Profiles, Roles, Permission Sets, OWD, Sharing Rules
- **E. Data Migration:** Salesforce Data Import Wizard
- **F. UI & Analytics:** Lightning App, Reports & Dashboards
- **G. Testing & Documentation:** Configuration testing, security testing, automation testing, and project documentation

**Thank You**